



Small nominalized clauses in Khalkha Mongolian: Asymmetries in mixed extended projections

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Abstract

This paper argues that just as there are SMALL NOMINALS (Pereltsvaig 2006, et seq.), there are also SMALL NOMINALIZED CLAUSES—nominalized clauses that lack a DP layer. Data come from four types of embedded clauses in Khalkha Mongolian (Khalkha) that differ from one another with respect to whether they are finite or nonfinite, nominalized or nonnominalized, and if nominalized, how much nominal structure is contained within them. Crucially, while complement nominalized clauses fully parallel possessive nominals in Khalkha in their nominal properties, converbial adjunct nominalized clauses display only a subset of these properties. Building on the idea that nominals may be smaller than DPs, I argue for an analysis that can account for the variation in nominal properties across the two nominalized clause types in Khalkha. I highlight a gap that emerges from this investigation—the absence of a small nominalized clause type containing a full CP structure nominalized by a reduced nominal structure that lacks DP. This suggests the existence of constraints on the construction of mixed extended projections. Building on a previous analysis by Svenonius (2021), I propose that the constraint can be understood in terms of the Williams Cycle (Williams 2003, et seq.) and related work.

Keywords Small nominalized clauses · Small nominals · Mixed extended projections · Nominalization · Williams Cycle · Khalkha Mongolian

1 Introduction

The mixed clausal and nominal properties of nominalized clauses (NCs) are often taken to reflect the presence of both clausal and nominal functional structure. Thus, NCs are said to contain a *mixed extended projection*, where nominal functional structure is projected immediately above clausal structure (Borsley and Kornfilt 2000; Kornfilt and Whitman 2011). While nominalization may take place at different levels of

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clausal structure (e.g., VP, vP, VoiceP, AspP, TP, CP) (see, e.g., Alexiadou 2001; Kornfilt and Whitman 2011), I will limit the discussion in this paper to NCs containing at least TP structure.¹

Mixed extended projections have been proposed in many analyses of NCs, such as in Modern Greek (Roussou 1991), Turkish (e.g., Borsley and Kornfilt 2000), Ndebele (Pietraszko 2019), Hill Mari (Voznesenskaia 2019), Khalkha Mongolian (Aravind 2021), Kazakh (Ótött-Kovács 2023), and Japanese (Nasu et al. 2024). The previous analyses are similar in that they propose the projection of a nominal layer above the clausal layer in NCs. With the exception of Nasu et al. 2024, previous analyses also uniformly analyze NCs as containing a DP projection. However, they differ in whether the clause is directly nominalized by DP (e.g., Roussou 1991; Borsley and Kornfilt 2000; Pietraszko 2019; Ótött-Kovács 2023), or whether the clause is first nominalized by n^0 (Aravind 2021)—a nominalizer in the sense of Marantz 2001 or a category-switching head in the sense of Rouveret 2023. Above nP, intermediate projections, such as PossP, may be projected before DP is projected (e.g., Voznesenskaia 2019). Recent work by Nasu et al. 2024, however, has argued that Japanese NCs are nominalized by n^0 , and that they may or may not contain a DP layer; in other words, Japanese contains NCs that only contain a nP layer. In sum, the previous literature has discussed three possible structures of NCs, schematized in (1).

- (1) Three analyses of mixed extended projections in NCs
 - a. NC directly nominalized by DP
DP > clausal structure
 - b. NC nominalized by nP and contains DP
DP > (...) nP > clausal structure
 - c. NC nominalized by nP and lacks DP
(no DP) (...) > nP > clausal structure

With this background in mind, this paper probes the internal structure of four types of embedded clauses in Khalkha Mongolian (henceforth Khalkha). I adopt a view of mixed extended projections in which the presence or absence of verbal or nominal properties is directly related to the absence or presence of dedicated functional projections (see, e.g., Alexiadou 2001; Alexiadou et al. 2010). Of the four embedded clause types, I show that three of them may host possessive enclitics, a property shared with possessive nominals in the language, which suggests that these clauses are NCs. However, only two of the three NCs—the argument NCs—fully parallel possessive nominals with respect to an additional set of nominal properties: the availability of genitive case assignment within the clause and the clause's ability to bear structural case marking. The remaining NC, a type of adjunct clause known as a converbial adjunct in the Mongolic literature (e.g., Janhunen 2012), does not display this additional set of nominal properties.

To account for the variation in nominal properties across Khalkha NCs, I argue that Khalkha provides evidence for two types of nominal structure in mixed extended projections: one that projects a full DP containing PossP and nP, and one that projects

¹The terms “clausal nominalization” and “propositional nominalization” have also been used to refer to NCs containing at least TP structure.

a smaller PossP containing nP but which lacks a DP layer. These two structures are schematized in (2). (2b) is similar to (1c), argued for by Nasu et al. (2024), in that both NCs lack a DP layer within the nominal superstructure.

(2) Two types of mixed extended projections in Khalkha

- a. Argument NC
DP > PossP > nP > clausal structure
- b. Converbial adjunct NC
PossP > nP > clausal structure

Building on Nasu et al. 2024, I propose that NCs may not always contain a DP layer. Crucially, I propose that clauses in Khalkha are first nominalized by nP, and that variation in nominal properties can be attributed to the presence or absence of dedicated nominal functional projections above nP. Without further stipulations, the direct-DP shell analysis in (1a) would not be able to account for why argument NCs in Khalkha display more nominal properties than converbial adjunct NCs, despite both showing evidence of being nominalized.

My analysis is also inspired by parallel observations in languages such as Tatar, where possessive nominals have been argued to vary as to whether they are full DPs or Small Nominals (SNs) that project PossP but lack DP (Pereltsvaig and Lyutikova 2014; Lyutikova and Pereltsvaig 2015). In this vein, I will refer to NCs that lack DP as *small nominalized clauses* (SNCs). As alluded to above, the evidence for a full DP-analysis of argument NCs and a SNC-analysis of converbial adjuncts in Khalkha is straightforward: argument NCs display the same set of nominal properties as possessive nominals, while converbial adjuncts display only a subset of those properties.

Below the level of nominalization, I also show that the four types of embedded clauses in Khalkha vary as to whether they are finite or nonfinite. Adopting the view that finiteness is an abstract property of clauses that may be inferred from a constellation of morphosyntactic properties (Kornfilt 2007; Nikolaeva 2007, 2010, 2012; McFadden and Sundaresan 2014; Rouveret 2023), I show that finite embedded clauses contain a nondefective T^0 and a full CP structure, while nonfinite clauses contain a defective T^0 and lack CP structure. This is schematized in (3):

(3) Finite and nonfinite embedded clauses in Khalkha

- a. Finite embedded clause
(nominal structure >) CP > $TP_{\text{nondefective}}$
- b. Nonfinite embedded clause
nominal structure > $TP_{\text{defective}}$

By the end of this paper, it will become clear that the four types of embedded clauses in Khalkha can each be uniquely understood as to whether they are finite or nonfinite, nominalized, or nonnominalized, and how much nominal functional structure they contain. We will see that nominalization crosscuts finiteness distinctions: both finite clauses containing a full CP and nonfinite clauses that lack CP may be nominalized.

Crucially, the landscape of NCs in Khalkha reveals an interesting gap. While Khalkha contains finite (i.e., full CP) clauses nominalized by a full DP and nonfinite (i.e., reduced TP) clauses nominalized by a full DP or reduced PossP, there is no

Table 1 Types of NCs in Khalkha

	PossP	DP
TP	Converbial adjunct	Nonfinite argument
CP	(Unattested)	Finite argument

evidence of a structure comprising a finite clause nominalized by a reduced PossP. I will refer to this unattested structure, represented by the bottom left cell in Table 1 below, as an example of an *improper mixed extended projection*.

The nonexistence of the bottom left cell in Table 1 represents the idea that a reduced nominal layer cannot nominalize a full(er) clausal structure—an idea that is not new. For example, Svenonius (2021) makes a similar but distinct prediction regarding the nonexistence of such structures. Building on his analysis, I show that the asymmetry in Table 1 can be understood in terms of the Williams Cycle (Williams 2003, et seq.) and related work.

The rest of this paper is organized as follows. Section 2 presents some background on Khalkha, and introduces the four types of embedded clauses examined in this paper. I propose a preliminary classification of these embedded clause types based on two surface morphosyntactic properties: the presence of overt tense markers and the presence of ϕ -agreeing possessive enclitics. In Sect. 3, I show that the four embedded clause types can be differentiated according to whether they are finite or nonfinite by further discussing two other morphosyntactic cues of (non)finiteness: whether T^0 is defective or nondefective and whether CP is present or absent in the embedded clause. In Sect. 4, I probe the amount of nominal structure present in embedded clauses by considering two other nominal properties: the presence of a genitive subject, and the presence of case marking on the clause. I show that variation in the nominal properties of the four embedded clause types can be understood in terms of variation in the amount of nominal structure they contain. In Sect. 5, I unify the findings from the previous sections to arrive at a micro-typology of embedded clause types in Khalkha. I propose to account for an unattested NC type by building on prior work by Svenonius (2021) and Williams (2003, 2011, 2013).

2 Background on Khalkha Mongolian

Khalkha Mongolian, or Khalkha, belongs to the Central Mongolic branch of the Mongolic language family, and refers to the standardized variety spoken in Mongolia. Some Khalkha data cited in this paper come from previous work (e.g., Guntsetseg 2016; Aravind 2021; Fong 2019; Gong 2022) and have been modified for consistency with the romanization and glossing conventions I adopt here. All uncited data come from my own fieldwork with four Khalkha consultants.²

This section presents an overview of the four embedded clause types in Khalkha that I investigate in this paper. I show two correlations between the surface morphology and size of the clause. I first discuss finite root clauses and possessive nominals

²While the number of consultants is relatively small, this is necessary for the in-depth fieldwork undertaken as part of this research. Future work will include consultations with a larger number of speakers (e.g., via a survey).

to introduce some relevant properties of finite clauses and nominals. Next, I discuss the four embedded clause types and sketch a preliminary classification of them based on their finiteness and nominal properties.

2.1 Finite root clauses

Khalkha is a head-final, SOV language with nominative-accusative case alignment and no overt subject-verb agreement. In finite root clauses, subjects surface with nominative case, which is morphologically unmarked, while definite and specific objects are typically marked with accusative case (see Guntsetseg 2016 for a detailed description of Khalkha differential object marking). For instance, in (4), the proper name object *Dorj* is obligatorily marked accusative.³

- (4) Tuyaa Dorj-iig üns-sen.
 T.NOM D.-ACC kiss-PST
 ‘Tuyaa kissed Dorj.’ (Guntsetseg 2016, 24, ex. 42)

As shown in (5), the order of morphemes within a basic verb complex is Verb Stem-Aspect-Negation-Tense. Notice that (5) contains the copula base *bai-* which hosts the nonpast tense suffix *-nA*. While I will not provide an analysis of the copula base here, descriptively, the appearance of the copula base seems to be related to a general restriction on having both an overt aspect suffix and an overt tense suffix co-occur on the same verb stem, at least in the formal, standardized variety of Khalkha.⁴ Given that Khalkha is uniformly head-final and assuming the Mirror Principle (Baker 1985), this implies that elements to the right are structurally higher and take scope over elements to the left. Thus, T^0 is higher than Neg^0 , which in turn is higher than Asp^0 and $Voice^0$. Following Peters 2024, I adopt the idea that external arguments are introduced by $Voice^0$ —a distinct head than v^0 , which verbalizes a base (e.g., Pylkkänen 2008; Harley 2013; Legate 2014). In line with the Uniformity of Theta Assignment Hypothesis (UTAH) (Baker 1988), I assume that the grammatical subject in Khalkha may originate as the external argument of a transitive or unergative verb in SpecVoiceP, or as the internal argument of an unaccusative verb from the complement position of V.⁵ Given that unaccusatives lack an external argument, I also assume that

³This paper adopts the Leipzig glossing rules with the following additions and modifications: C complementizer, C.NML nominalized complementizer, CCMT.CVB concomitant converb, COND.CVB conditional converb, EXCL exclamative, HAB habitive, NPST nonpast, PST.IMPF past imperfect, RP reflexive possessive, SUCC.CVB successive converb, TEMP.CVB temporal converb, TERM.CVB terminative converb.

I use uppercase A to represent vowels that undergo a four-way vowel harmony: $a \sim e \sim o \sim \ddot{o}$, and uppercase U to represent vowels that undergo a two-way vowel harmony: $u \sim \ddot{u}$.

⁴In the colloquial register of Khalkha, it is possible to pronounce both overt aspect and overt tense suffixes together without *bai-*insertion, as seen (i). However, I do not know whether this is possible for all combinations of overt aspect and overt tense.

- (i) Odgerel huushuur hii-gee-güi-n.
 O.NOM khuushuur do-IPFV-NEG-NPST
 ‘Odgerel hasn’t made khuushuur.’

⁵To the best of my knowledge, there are no reliable diagnostics in Khalkha to distinguish unaccusative subjects from unergative ones.

VoiceP is not projected in such structures. I provide evidence in Sect. 3 that subjects obligatorily raise to SpecTP.

- (5) Khalkha verb complex: $V^0 < Asp^0 < Neg^0 < T^0$

Odgerel khuushuur hii-gee-güi bai-na.⁶

O.NOM khuushuur do-IPFV-NEG COP-NPST

‘Odgerel hasn’t made khuushuur (fried pastry with meat filling).’

2.2 Possessive nominals

As I will show later, NCs share many morphosyntactic properties with possessive nominals in Khalkha. Thus, it will be worthwhile to go over some properties of possessive nominals. There are three ways to express possession in Khalkha: via a genitive-marked possessor argument (6a), a ϕ -agreeing possessive morpheme (POSS) which references the possessor argument (6b), or both (6c). Like in many Mongolic languages, rather than case attaching outside possessive morphology, possessive morphology attaches outside case (see, e.g., Janhunen 2003; Gong 2021). Thus, I will assume that POSS is an enclitic whose surface position is postsyntactically derived, although I will not provide an analysis here. See Gong 2021, however, for a Lowering account that derives POSS-final morpheme order in the Mongolic language Dagur.

- (6) Three ways to express possession in Khalkha

- a. Genitive pronoun

Chi minii nom-iig unsh-san.

2SG.NOM 1SG.GEN book-ACC read-PST

- b. Possessive enclitic

Chi nom-iig=*min*’ unsh-san.

2SG.NOM book-ACC=1SG.POSS read-PST

- c. Genitive pronoun + possessive enclitic

Chi minii nom-iig=*min*’ unsh-san.

2SG.NOM 1SG.GEN book-ACC=1SG.POSS read-PST

‘You read my book.’

While the acceptability of possessive clitic doubling structures, such as (6c) is debated, it has been documented by various authors and is accepted by my consultants under an emphatic reading of the possessor (see also Hammar 1983; Brosig

⁶Tense suffixes are not always overt in Khalkha and further research is required to understand the precise distribution of null and overt forms. Consider (i), which is given a translation in the past tense, but does not contain any of the overt past tense suffixes (see Table 3). For concreteness, I will assume that there are null variants of the nonpast and past tense suffixes, although nothing crucial hinges on this assumption.

- (i) Tüünii nom, devter ol-d-son-güi-Ø.

3SG.GEN book notebook find-PASS-PFV-NEG-PST

‘His books and notebooks were not found.’

(Tserenpil and Kullmann 2008, 149)

2.3 Embedded clauses

While argument clauses may serve as clausal subjects or complements, I will focus only on complement clauses in this paper. The three types of complement clauses I investigate in this paper are the following: finite bare CPs (which are nonnominalized) (7), finite complement NCs (8), and nonfinite complement NCs (9). In (7)–(9), I have provided all possible cases that the embedded subject may surface with in each clause type. Consider first whether the complement clauses in (7)–(9) may contain an overt tense suffix, such as the nonpast suffix *-nA*. While (7) and (8) may contain *-nA*, (9b) may not. Instead the so-called futurity/infinite aspect marker *-h* (Janhunen 2012, 160) is used in (9a) to express the future reading. Whether the complement clause may contain an overt tense suffix (e.g., *-nA*) appears to correlate with the presence of an overt complementizer *gej* or *ged(e)g*. Given that the expression of overt tense marking⁹ is often tied to finiteness (e.g., Chomsky 1980, 2000, 2001), I will refer to clauses like (7) and (8) as “finite,” and clauses like (9a) as “nonfinite.”¹⁰

'You killed my only daughter.' (example from MNC; retrieved 2022/06/09)

¹⁰I am using the terms “finite” and “nonfinite” slightly differently than it is used in the Mongolic literature. In that literature, “finite” and “nonfinite” are terms that describe types of verb inflections and the clauses that those verb inflections may appear in. Verb endings that are “finite” are negatively defined as being unable to appear in dependent clauses, such as certain types of NCs, converbial adjuncts, and relative clauses. This accords with the definition of finiteness I adopt, in that finite tense suffixes are

- (7) Finite bare CP may contain nonpast *-nA*
 Bi [Naraa-g/Ø margaash ir-*ne* gej] mart-san.
 1SG.NOM N.-ACC/NOM tomorrow come-NPST C forget-PST
 ‘I forgot that Naraa will come tomorrow.’
- (8) Finite complement NC may contain nonpast *-nA*
 Bi [Naraa-g/giin/Ø margaash ir-*ne* gedg]-iig
 1SG.NOM N.-ACC/GEN/NOM tomorrow come-NPST C.NML-ACC
 mart-san.
 forget-PST
 ‘I forgot that Naraa will come tomorrow.’
- (9) Nonfinite complement NC cannot contain nonpast *-nA*
 a. Bi [Naraa-g/giin/Ø margaash ire-*h*]-iig mart-san.
 1SG.NOM N.-ACC/GEN/NOM tomorrow come-INF-ACC forget-PST
 ‘I forgot that Naraa is coming tomorrow.’
 b. *Bi [Naraa-g/giin/Ø margaash ire-*ne*]-g mart-san.
 1SG.NOM N.-ACC/GEN/NOM tomorrow come-NPST-ACC forget-PST
 Int. ‘I forgot that Naraa is coming tomorrow.’

The variation in complementizer forms can be traced to the fact that *ge* is historically the root of the verb ‘to say’; in modern usage, it has mostly been replaced by *hel* ‘to say.’ As evidence of their verbal provenance, it is possible to deconstruct *gej* as comprising the root *ge* + the converb *-j*, and *ged(e)g* as comprising *ge* + the habitive aspect suffix *-d(e)g*. In fact, as a reviewer notes, other complementizer forms involving different aspect suffixes are attested, such as *ge* + the futuritive/infinite suffix *-h* and *ge* + the perfective suffix *-s(e)n*. *Ge* itself may still be used as a verb, for example, *gene* ‘say(s); will say,’ which comprises the root *ge* + the nonpast tense suffix *-ne*. Further, forms such as *gene* cannot co-occur with another ‘say’-verb, implying that it is not used as a complementizer.¹¹ Different forms of the complementizer are associated with slightly different meanings regarding the nature of the information reported in the embedded clause, and an in-depth investigation of these meanings is beyond the scope of this paper.¹² However, many authors note that

indeed only found in finite clauses. Conversely, verb endings that may appear in dependent clauses are described as “nonfinite,” even though “nonfinite” endings may themselves appear in “finite” clauses that permit the use of “finite” verb endings. This distributional definition of finiteness differs from that in the theoretical syntactic literature, where finiteness refers to an abstract and structural property of clauses tied to the properties of functional heads (e.g., of T^0) and clause size (e.g., whether a full CP is projected; see, e.g., Kornfilt 2007; Nikolaeva 2010, 2012; McFadden and Sundaresan 2014; Rouveret 2023; and the collection of articles in Nikolaeva 2007). It is this latter notion of finiteness that I adopt in this paper.

¹¹ A reviewer raises the possibility that *gej* clauses could be analyzed as adjuncts (Major 2024). However, evidence from Fong 2019 argues against this view: subjects in *gej* clauses can undergo hyperraising into the matrix clause, which is unexpected if such clauses were adjuncts.

¹² For example, one consultant suggests that *ges(e)n* is used when the information reported in the embedded clause is secondhand information (see also Tserenpil and Kullmann 2008, 309). Another consultant, however, does not share this intuition, and reports that *ged(e)g* and *ges(e)n* can be used interchangeably.

gej and *ged(e)g* are most commonly used (see, e.g., Tserenpil and Kullmann 2008, 309, Janhunen 2012, p. 283–284 and Gong 2022, p. 24–27). This is consistent with how my consultants use them. Thus, I adopt the view that *gej* and *ged(e)g* have been grammaticalized as complementizers in modern Khalkha. Specifically, I will treat *ged(e)g* as an allomorph of *gej* in the context of nominalizing n^0 , and I gloss it as C.NML.

(10) Vocabulary Items for C^0 in Khalkha

- a. $C^0 \leftrightarrow gej$
- b. $C^0 \leftrightarrow ged(e)g / __ n^0$

Next, consider whether the embedded complement clause may host a ϕ -agreeing possessive enclitic, as seen on possessive nominals in (6). Both complement NCs (12) and (13), which are marked with accusative case, may optionally host possessive enclitics. It is not possible for finite bare CPs, such as (11), to host possessive enclitics; notice as well that it is not marked with any case (hence I describe it as “bare”). For simplicity, I only represent accusative subjects in (11)–(13).¹³

(11) Finite bare CP may not host a possessive enclitic

Bi [Naraa-g unta-j bai-jee gej](=*n') med-ne.
 1SG.NOM N.-ACC sleep-PROG COP-PST.PRF C=3.POSS know-NPST
 ‘I know that Naraa was sleeping.’

(12) Finite complement NC may host a possessive enclitic

Bi [Naraa-g unta-j bai-jee gedg]-iig(=n')
 1SG.NOM N.-ACC sleep-PROG COP-PST.PRF C.NML-ACC=3.POSS
 med-ne.
 know-NPST
 ‘I know that Naraa was sleeping.’

(13) Nonfinite complement NC may host a possessive enclitic

Bi [Naraa-g unta-j bai-sn]-iig(=n') med-ne.
 1SG.NOM N.-ACC sleep-PROG COP-PFV-ACC=3.POSS know-NPST
 ‘I know that Naraa was sleeping.’

Turning next to converbial adjuncts, these are subordinate adjuncts (Haspelmath 1995) that modify the main clause. While many converbial adjuncts function as temporal modifiers, there are also ones that have nontemporal functions, e.g., conditionals. Converbial adjuncts contain a class of verb inflections called ‘converb markers’ (Janhunen 2012, 163) that I will argue to be nonfinite T^0 . For ease of reference, I will simply refer to these inflections as ‘converbs.’ Two examples of con-

¹³Embedded clauses containing a ϕ -agreeing possessive enclitic are found by my consultants to be acceptable with accusative and genitive subjects, while nominative subjects are judged to be degraded. Given that accusative and genitive subjects are thought to occupy a higher position within the embedded clause left periphery than nominative subjects (e.g., Kornfilt and Whitman 2012; Baker and Vinokurova 2010; Aravind 2021; but see Miyagawa 2011), I hypothesize that ϕ -agreement resulting in the exponence of the possessive enclitics is only possible when the subject has moved to a higher position where it can also be assigned accusative or genitive case. Further research is required to understand the relationship between subject case and possessive enclitics; see Lim 2024a for some discussion.

verbial adjuncts are shown below: (14a) contains a general temporal converb *-hAd* (glossed TEMP.CVB), which indicates simultaneity or cotemporality; (14b) contains the concomitant converb *-ngUUt* (glossed CCMT.CVB), which is used to express an action/event that usually accompanies or precedes the one in the main clause. The subjects of converbial adjuncts may be accusative or nominative.

(14) Converbial adjuncts

- a. Bi [tüüniig/ter hicheel-Ø=ee hii-j
1SG.NOM 3SG.ACC/3SG.NOM lesson-ACC=RP do-PROG
bai-had] unt-san.¹⁴
COP-TEMP.CVB sleep-PST
'When she/he was studying, I fell asleep.'
- b. Bi [Naraa-g/Ø unta-nguut] video togloom toglo-son.
1SG.NOM N.-ACC/NOM sleep-CCMT.CVB video game play-PST
'As soon as Naraa fell asleep, I played video games.'

It is not possible to use an overt tense marker, such as the nonpast *-nA*, in converbial adjuncts, as shown in (15). The future tense interpretation of the converbial adjunct is evident from the use of the adverb *margaash* 'tomorrow,' which follows the embedded subject *Naraa* in (15). The use of a converb thus appears to be in complementary distribution with the use of an overt tense suffix. Notice further that converbial adjuncts, like nonfinite NCs (e.g., (9)), do not contain the overt complementizer *gej* that is found in finite embedded clauses (e.g., (7) and (8)). Since they do not allow overt expression of finite tense, I will treat converbial adjuncts as nonfinite clauses.

(15) *-hAd* converbial adjunct cannot contain nonpast *-nA*

- Bi [Naraa margaash {ire-hed / *ir-ne
1SG.NOM N.NOM tomorrow come-TEMP.CVB come-NPST
bai-had}] huushuur hii-ne.
COP-TEMP.CVB khuushuur do-NPST
'I will make khuushuur when Naraa comes tomorrow.'

Finally, observe that converbial adjuncts may optionally host ϕ -agreeing possessive enclitics, as seen in (16). Only accusative subjects are represented in (16) (see footnote 13). Since the ability to host possessive enclitics is a property of possessive nominals in Khalkha, this suggests that converbial adjuncts contain some nominal structure (i.e., they are nominalized clauses).¹⁵ However, I will argue in Sect. 4 that converbial adjuncts and argument NCs do not contain the same amount of nominal structure.

¹⁴With the exception of kinship and friendship terms, the accusative case suffix on nouns is commonly realized as zero in the context of the reflexive possessive (RP) enclitic. See Guntsetseg 2016, 58–59, 82–83 for some discussion.

¹⁵Diachronically, converbial adjuncts in Khalkha are argued to have developed from case-marked forms of verbal constituents such as participial or gerundive phrases (e.g., Ramstedt 1903; Poppe 1954). However, I do not assume that their origins reflect their underlying structure synchronically (cf. Umetani 2019).

Table 2 Morphosyntactic properties of Khalkha embedded clauses

	Clause type			
	Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
Able to contain overt tense suffixes	✓	✓	✗	✗
Able to host possessive suffixes	✗	✓	✓	✓

(16) Converbial adjuncts may host possessive enclitics

- a. Bi [tüüniig hicheel-Ø=ee hii-j
 1SG.NOM 3SG.ACC lesson-ACC=RP do-PROG
 bai-had](=n') unt-san.
 COP-TEMP.CVB=3.POSS sleep-PST
 ‘When she/he was studying, I fell asleep.’
- b. Bi [Naraa-g unta-nguut](=n') video togloom toglo-son.
 1SG.NOM N.-ACC sleep-CCMT.CVB=3.POSS video game play-PST
 ‘As soon as Naraa fell asleep, I played video games.’

2.4 Summary

To recap the findings above, we have seen that the four embedded clause types in Khalkha may be differentiated along two dimensions. First, the embedded clause types may be differentiated according to whether they are finite or nonfinite, diagnosed by their ability to contain an overt tense ending. Finite bare CPs, and finite NCs containing the overt complementizer *gej/ged(e)g* may contain overt tense markers, while nonfinite NCs and converbial adjuncts may not. Second, the embedded clause types may be differentiated according to whether they are nominalized or non-nominalized, diagnosed by their ability to host a ϕ -agreeing possessive enclitic. Finite NCs, nonfinite NCs and converbial adjuncts may do so, but finite bare CPs may not. Table 2 summarizes these morphosyntactic properties of the four embedded clause types.

In addition to differential object marking, Khalkha also displays differential subject marking, where the subject of an embedded clause may surface with accusative, genitive, or (morphologically unmarked) nominative case (see Guntsetseg 2016 for discussion). Many authors have argued for treating accusative case on embedded subjects in Khalkha as a dependent case under a configurational view of case assignment (Aravind 2021; Gong 2023a,b; Lim 2022; Peters 2020, 2024).¹⁶ In the rest of this paper, I will omit accusative subjects in the examples unless necessary, and focus on illustrating embedded clauses with genitive and nominative (case-unmarked) subjects. There are two main reasons for this. First, I show that the distribution of nominative (case-unmarked) subjects is not sensitive to whether the clause is finite or nonfinite, a finding that can be significant for understanding how nominative case is assigned in Khalkha. Second, genitive subjects display an interesting distribution with respect

¹⁶See, however, Fong 2019 for a functional head account of accusative embedded subjects, with accusative case being assigned by matrix transitive v^0 .

to other nominal properties, such as case marking on the embedded clause and the presence of ϕ -agreeing possessive enclitics. The presence of genitive subjects is thus instructive as to amount and type of nominal functional structure present in an embedded clause, and can be significant for understanding how genitive case is assigned in Khalkha.

In the next section, I will further probe the structure of embedded clauses below the level of nominalization (if any), and investigate its connection with finiteness and nonfiniteness.

3 (Non)finiteness and clausal structure in embedded clauses

As mentioned in the introduction, this paper argues for two types of mixed extended projections in Khalkha, repeated in (2). In this section, I will concentrate on investigating the clausal structure below the level of nominalization in the four types of embedded clauses discussed in this paper.

(2) Two types of mixed extended projections in Khalkha

- a. Argument NC
DP > PossP > nP > *clausal structure*
- b. Converbial adjunct NC
PossP > nP > *clausal structure*

I will show that in Khalkha, finite clauses contain a full CP as well as TP, while nonfinite clauses contain TP but lack CP. Put differently, finite clauses contain more clausal structure than nonfinite ones. This is schematically represented in (17).¹⁷

(17) Clausal structure of embedded clauses in Khalkha

- a. Finite clauses (finite bare CP, finite NC)
(nominal structure) > CP > TP
- b. Nonfinite clauses (nonfinite NC, converbial adjunct)
nominal structure > TP

I will first discuss evidence showing that TP is projected in all of the embedded clauses discussed here. This will narrow the space of variation to i) the type of T^0 (defective vs. nondefective) contained in embedded clauses; and ii) whether embedded clauses contain CP structure. I show that embedded clauses which may contain overt tense (i.e., finite bare CPs and finite NCs) contain a nondefective flavor of T^0 and a full CP structure. On the other hand, clauses which may not contain overt tense (i.e., nonfinite NCs and converbial adjuncts) contain a defective flavor of T^0 and lack CP structure. I end this section with a discussion of how the findings can be significant for understanding how nominative case is assigned in Khalkha.

¹⁷I leave open the possibility that there may be additional structure above TP within the CP domain, which is consistent with the view that clauses contain a more articulated left periphery (e.g., Rizzi 1997). This does not affect the main argument: that nonfinite clauses contain less clausal structure than finite clauses.

3.1 The presence of TP

I argue that the four embedded clause types under consideration in Khalkha contain at least a TP layer. To support my argument, I show that subjects in the four embedded clause types move to SpecTP, based on i) scope effects and ii) the placement of adverbs.

Consider first the interpretation of subjects with respect to clausal negation. Since Khalkha is head-final, other diagnostics besides word order must be used to determine whether the external argument has raised to SpecTP. Adopting an analysis by Miyagawa (2001) for Japanese, I show that external arguments obligatorily scope over negation in Khalkha, suggesting that they move to a specifier position above NegP, which I take to be SpecTP.

In the Japanese sentence (18), that T^0 -*ta* PST is higher than Neg⁰ -*nakat*, which in turn is higher than the verb root *home* ‘to praise,’ is indicated by the morpheme order within the verb complex: $V^0 < \text{Neg}^0 < T^0$. In (18), the universal quantifier direct object can only be interpreted within the scope of negation.

- (18) Taroo-ga *zen'in-o* *home-nakat-ta* (yo / to *omou*).
 Taro-NOM all-ACC praise-NEG-PST EXCL C think
 ‘(I think that) Taro didn’t praise all (!).’ $\checkmark \neg > \forall, (*)\forall > \neg$ ¹⁸
 (Miyagawa 2001, 298, ex. 7)

Miyagawa (2001) argues that if external arguments A-move to SpecTP in Japanese, then a universal quantifier subject should not be interpreted within the scope of negation. This prediction is borne out as seen in (19).

- (19) *Zen'in-ga* *sono tesuto-o* *uke-nakat-ta* (yo / to *omou*).
 all-NOM that test-ACC take-NEG-PST EXCL C think
 ‘(I think that) All did not take that test (!).’ $*\neg > \forall, \checkmark \forall > \neg$
 (Miyagawa 2001, 298, ex. 8)

Returning to Khalkha, in the root clause (20), we see that a universally quantified direct object has to be interpreted within the scope of negation. This is expected as the direct object *büh huushuur* ‘all khuushuur’ occupies the complement of V^0 , where it is c-commanded by Neg⁰. (21) shows that a universally quantified subject must take scope outside of negation (note that the verb *tents-eh* ‘to pass’ assigns dative case to its internal argument). This suggests that the subject *büh oyutuud* ‘all students’ must be located in a position above NegP, where it is not c-commanded by Neg⁰. A natural candidate for this landing site above NegP is SpecTP.¹⁹

¹⁸Miyagawa (2001) uses (*) to indicate that some speakers accept the $\forall > \neg$ reading, which he attributes to a collective reading of ‘all.’ The fact that such an interpretation is unavailable in Khalkha (20) suggests that *büh* ‘all’ does not possess a collective reading.

¹⁹Strictly speaking, it is not logically possible to construct an $\forall > \neg$ scenario that is incompatible with a $\neg > \forall$ reading, as every scenario in which all students did not pass the exam (verifying $\forall > \neg$) is also a scenario in which not all students passed the exam (verifying $\neg > \forall$). It would appear that the weaker reading—the reading that is logically entailed by the stronger reading which matches the scope relation between the quantifier and negation—is dispreferred by my consultants, perhaps due to pragmatic competition with the stronger reading. I thank a reviewer for their insightful comments regarding this point.

- (20) Bat *büh huushuur-iig* id-ee-*güi* bai-na
 B.NOM all khuushuur-ACC eat-IPFV-NEG COP-NPST
 ‘Bat did not eat all the khuushuur.’
 #Context: There are 10 khuushuur and Bat didn’t eat any. $*\forall > \neg$
 ✓Context: There are 10 khuushuur and Bat ate one. $\checkmark \neg > \forall$
- (21) *Büh oyutn-uud* ter shalgalt-and tents-ee-*güi* bai-na
 all student-PL.NOM that exam-DAT pass-IPFV-NEG COP-NPST
 ‘All the students didn’t pass that exam.’
 ✓Context: There are 10 students in the class and no one passed that exam. $\checkmark \forall > \neg$
 #Context: There are 10 students in the class and only two passed that exam. $*\neg > \forall$

The same scope effects can be shown for universally quantified subjects in finite bare CPs (22), finite NCs (23), nonfinite NCs (24) and converbial adjuncts (25)–(26). Looking at the complement clauses first, (22)–(24) show that the embedded subject *büh oyutnuud* ‘all students’ always takes scope above embedded negation.

- (22) Universally quantified subject scopes over negation in finite bare CP
 Bilgüün [*büh oyutn-uud* ter shalgalt-and tents-ee-*güi* bai-na
 B.NOM all student-PL.NOM that exam-DAT pass-IPFV-NEG COP-NPST
 gej] hel-sen.
 C say-PST
 ‘Bilguun said that all students didn’t pass that exam.’
 ✓Context: There are 10 students in the class and no one passed that exam. $\checkmark \forall > \neg$
 #Context: There are 10 students in the class and only two passed that exam. $*\neg > \forall$
- (23) Universally quantified subject scopes over negation in finite NC
 Bilgüün [*büh oyutn-uud* ter shalgalt-and tents-ee-*güi* bai-na
 B.NOM all student-PL.NOM that exam-DAT pass-IPFV-NEG COP-NPST
 gedg]-iig hel-sen.
 C.NML-ACC say-PST
 ‘Bilguun said that all students didn’t pass that exam.’
 ✓Context: There are 10 students in the class and no one passed that exam. $\checkmark \forall > \neg$
 #Context: There are 10 students in the class and only two passed that exam. $*\neg > \forall$

- (24) Universally quantified subject scopes over negation in nonfinite NC

Bilgüün [*büh oyutn-uud* ter shalgalt-and tents-ee-güi]-g
 B.NOM all student-PL.NOM that exam-DAT pass-IPFV-NEG-ACC
 hel-sen.
 say-PST

‘Bilguun said that all students didn’t pass that exam.’

✓Context: There are 10 students in the class and no one passed that exam.

✓∀ > ¬

#Context: There are 10 students in the class and only two passed that exam.

*¬ > ∀

(25) and (26) show adjuncts headed by the conditional converb *-vAl* (glossed COND.CVB) and the temporal converb *-hAd* (glossed TEMP.CVB), respectively. These sentences, while judged to be grammatically acceptable, have pragmatically strange readings as the embedded subject *büh oyutnuud* ‘all students’ take scope above embedded negation within each adjunct clause. In (25), the reading that the teacher will be happy if all their students fail the exam obtains, while in (26), the reading that obtains is the one where the teacher explained when all students did not understand, instead of the one where at least some students did not understand.

- (25) Universally quantified subject scopes over negation in *-vAl* converbial adjunct

Bagsh [*büh oyutn-uud* ter shalgalt-and tents-ee-güi
 teacher.NOM all student-PL.NOM that exam-DAT pass-IPFV-NEG
 bai-val] bayarla-h bol-no.
 COP-COND.CVB be.happy-INF become-NPST

‘The teacher will be happy if all students do not pass the exam.’

✓Context: The teacher is unkind and wants all students to fail the exam.

✓∀ > ¬

#Context: The teacher has low expectations of their students, and will be happy if at least one student passes the exam.

*¬ > ∀

- (26) Universally quantified subject scopes over negation in *-hAd* converbial adjunct

Bagsh [*büh oyutn-uud* asuult-iig oilgo-h-güi
 teacher.NOM all student-PL.NOM question-ACC understand-INF-NEG
 bai-had] tailbarla-san.
 COP-TEMP.CVB explain-PST

‘The teacher explained when all the students did not understand the question.’

✓Context: In a class of 10, the teacher explained when all 10 students did not understand the question.

✓∀ > ¬

#Context: In a class of 10, the teacher explained when not all 10 students understood the question.

*¬ > ∀

In sum, the examples above show that universally quantified subjects are interpreted outside the scope of negation, supporting the idea that subjects move to SpecTP in all four embedded clause types discussed here.

Next, consider the placements of manner, aspectual, and temporal adverbs in Khalkha, which I take to adjoin to vP, AspP, and TP, respectively. First, Guntsetseg (2016) shows that the adverb *dandaa* ‘always’—an AspP adverb—may immediately precede the verb or the dative-marked internal argument of *tusal-ah* ‘to help’ (which assigns dative case to its internal argument). However, it cannot appear to the left of the external argument subject, as shown (27). Given that external arguments are introduced in SpecVoiceP, which is below AspP, the inability of *dandaa* to precede an external argument subject can be explained by the fact that subjects move to SpecTP. On this view, since the subject is located in SpecTP, it cannot appear to the right of *dandaa*. In (27), I take the order *dandaa* > DO to reflect the base order, while DO > *dandaa* may be derived via object scrambling to a position below SpecTP.

- (27) { *Dandaa } Tuyaa { *dandaa* } Dorj-id { *dandaa* } tusal-dag.
 always T.NOM always D.-DAT always help-HAB
 ‘Tuyaa always helps Dorj.’ (Guntsetseg 2016, 25, ex. 46)

Given my account which takes external arguments to be introduced in SpecVoiceP, which is directly above vP, it is unsurprising that a manner adverb, such as *hurdan* ‘quickly’—which adjoins to vP—may not precede an external argument subject as shown in (28). The placement of manner adverbs with respect to external argument subjects is thus uninformative as to whether subjects move to SpecTP.²⁰

- (28) { *Hurdan } Bilgüün { *hurdan* } ene baishin-g { *hurdan* } bari-san.
 quickly B.NOM quickly this building-ACC quickly build-PST
 ‘Bilguun built this building quickly.’

However, if SpecTP is a derived position, we might expect that nonagentive arguments may also raise to satisfy the EPP feature on T⁰. In passive constructions, the external argument is absent and the internal argument of the verb functions as the theme subject. If subjects raise to SpecTP to check T⁰’s EPP feature, we predict that a manner adverb, such as *hurdan*, which adjoined to vP, should not be able to precede a passivized, theme subject in SpecTP. This prediction is borne out, as seen in (29).

²⁰Interestingly, Gong (2023a) reports that examples such as (ib), where a manner adverb precedes an external argument subject, are acceptable:

- (i) a. Bat *hurdan* güi-j bai-na.
 B.NOM quickly run-PROG COP-NPST
 ‘Bat is running quickly.’
 b. *Hurdan*₁ Bat —₁ güi-j bai-na.
 quickly B.NOM run-PROG COP-NPST
 ‘Bat is running quickly.’ (Gong 2023a, 417, fn.2)

Gong suggests that *hurdan* ‘quickly’ is a VP-level adverb and that it may undergo clause-internal scrambling, deriving (ib). Such data contradict the judgements of my consultants (e.g., (28)) as well as those previously reported in Guntsetseg 2016; Fong 2019; Peters 2024. I assume that Khalkha speakers may vary as to whether they accept clause-internal scrambling of low adverbs.

- (29) {*Hurdan} ene baishin {hurdan} bari-gd-san.
 quickly this building.NOM quickly build-PASS-PST
 ‘This building was built quickly.’

Compare the passive sentence in (29) with its active counterpart in (28). In (28), the manner adverb *hurdan* may immediately precede the verb or the accusative-marked theme object but not the agentive subject. However, in (29), while *hurdan* may precede the verb, it may not precede the theme subject. This suggests that in the absence of an external argument, the theme argument raises to SpecTP to satisfy the EPP feature on T⁰. The high position of the theme argument in passives thus reinforces the idea that subjects move to SpecTP and that SpecTP is a derived position.

While aspectual and manner adverbs are useful diagnostics for subject position, temporal adverbs, such as *margaash* ‘tomorrow,’ are less reliable in this regard as they display more flexibility in their distribution. For instance, *margaash* may appear before or after the clausal subject, as shown in (30).

- (30) {Margaash} Naraa {margaash} ene baishin-g nuraa-na.
 tomorrow N.NOM tomorrow this building-ACC tear.down-NPST
 ‘Naraa will tear down this building tomorrow.’

I assume that temporal adverbs adjoin to TP, and that the order in which *margaash* precedes a case-unmarked subject reflects the base order, while the order in which a case-unmarked subject precedes *margaash* is derived via short scrambling of the subject to a position above TP.²¹

Next, consider the distribution of manner adverbs in the four types of embedded clauses in Khalkha. First, Fong (2019) shows that finite bare CPs may contain a temporal adverb, such as *margaash* ‘tomorrow,’ which is unambiguously interpreted as modifying the embedded clause (31). This suggests that finite bare CPs contain a TP layer.

- (31) Finite bare CPs may contain temporal adverbs
 Bat [margaash Dulmaa nom unshi-na gej] hel-sen.
 B.NOM tomorrow D.NOM book read-NPST C say-PST
 ‘Bat said that Dulmaa will read a book tomorrow.’ (Fong 2019, 2, ex. 3)

Fong (2019) also shows that manner adverbs, such as *hurdan* ‘quickly,’ cannot precede a case-unmarked subject (32). However, given that manner adverbs adjoin to vP and do not scramble higher (however, see footnote 20 for conflicting judgements reported by Gong 2023a), it is expected that they cannot precede an external argument subject, which I take to be introduced in SpecVoiceP.

²¹ In fact, temporal adverbs may also precede accusative and genitive subjects, which many works analyze as having moved to higher positions within the clausal periphery (see, e.g., Fong 2019; Aravind 2021; Peters 2024; Lim 2025). This suggests that temporal adverbs themselves may also scramble to the edge of the embedded (nominalized) clause.

- (32) Manner adverbs cannot precede subjects in finite bare CPs

Dorj [[**hurdan*] Naraa {*hurdan*} baishin bari-san ge:] hel-sen.
 D.NOM quickly N.NOM quickly house build-PST C say-PST
 ‘Dorj said that Naraa quickly built a house.’ (Fong 2019, 16, ex. 48)

Nevertheless, it is also not possible for a manner adverb to precede a theme subject in a finite bare CP, as shown in (33). Since the theme subject is first introduced in the complement position of V^0 , the fact that it cannot follow *hurdan* suggests that it has raised to SpecTP within the embedded clause.

- (33) Manner adverb cannot precede theme subject in finite bare CP

Dorj [[**hurdan*] ene baishin {*hurdan*} bari-gd-san ge:]
 D.NOM quickly this building.NOM quickly build-PASS-PST C
 hel-sen.
 say-PST
 ‘Dorj said that this building was built quickly.’

Next, I show that the same adverb distribution facts also hold in finite NCs, non-finite complement NCs and converbial adjuncts. Consider the pair of finite NCs in (34) and (35). (34) shows that a finite NC may contain a temporal adverb *margaash* ‘tomorrow,’ suggesting the presence of TP in the embedded clause. (35) shows that a theme subject may not follow the manner adverb *hurdan* ‘quickly,’ suggesting that the embedded subject has moved to SpecTP.

- (34) Finite NC may contain temporal adverbs

Dorj [*margaash* Naraa tögöldör huur toglo-no gedg]-iig hel-sen.
 D.NOM tomorrow N.NOM piano play-NPST C.NML-ACC say-PST
 ‘Dorj said that Naraa will play the piano tomorrow.’

- (35) Manner adverb cannot precede theme subject in finite NC

Dorj [[**hurdan*] ene baishin {*hurdan*} bari-gd-san
 D.NOM quickly this building.NOM quickly build-PASS-PST
 gedg]-iig sons-son.
 C.NML-ACC hear-PST
 ‘Dorj heard that this building was built quickly.’

In (36), the nonfinite NC may contain the temporal adverb *margaash* ‘tomorrow,’ suggesting the presence of TP within the embedded clause. In (37), the theme subject may not follow the manner adverb *hurdan* ‘quickly,’ suggesting that the embedded subject has moved to SpecTP.

- (36) Nonfinite NCs may contain temporal adverbs

Dorj [*margaash* Naraa ene baishin-g nuraa-h]-iig med-sen.
 D.NOM tomorrow N.NOM this building-ACC tear.down-INF-ACC know-PST
 ‘Dorj knew that tomorrow Naraa will tear down this building.’

- (37) Manner adverb cannot precede theme subject in nonfinite NC

Dorj [{*hurdan} ene baishin {hurdan} bari-gd-sn]-iig
 D.NOM quickly this building.NOM quickly build-PASS-PFV-ACC
 med-sen.
 know-PST
 ‘Dorj knew that this building was built quickly.’

Turning to converbial adjuncts, (38) shows that a converbial adjunct may contain the temporal adverb *margaash* ‘tomorrow,’ given that the case-unmarked embedded subject may precede *margaash* (which I assume is due to the availability of scrambling). (39) shows that the theme subject may not follow the manner adverb *hurdan* ‘quickly,’ suggesting that the embedded subject has moved to SpecTP.

- (38) Temporal adverbs may precede subjects in converbial adjuncts

Bi [{Naraa} *margaash* {Naraa} ir-tel] huushuur
 1SG.NOM N.NOM tomorrow N.NOM come-TERM.CVB khuushuur
 hii-h-güi.
 do-INF-NEG
 ‘I will not make khuushuur until Naraa comes tomorrow.’

- (39) Manner adverb cannot precede theme subject in converbial adjunct

Bilgüün [{*hurdan} ene nom {hurdan} ol-d-vol]
 B.NOM quickly this book.NOM quickly find-PASS-COND.CVB
 uurla-h-güi.
 be.angry-INF-NEG
 ‘If this book is found quickly, Bilguun will not be angry.’

Finally, consider the fact the four embedded clause types may contain an independent temporal adverb that mismatches in tense interpretation with another temporal adverb located in the matrix clause. This is shown in (40)–(43), and provides further evidence for TP structure in the four embedded clause types.

- (40) Temporal mismatch in finite bare CP

Öchigdör bi [*margaash* Naraa ir-ne gej] hel-sen.
 yesterday 1SG.NOM tomorrow N.NOM come-NPST C say-PST
 ‘Yesterday I said that Naraa will come tomorrow.’

- (41) Temporal mismatch in finite NC

Öchigdör bi [*margaash* Naraa ir-ne gedg]-iig mart-san.
 yesterday 1SG.NOM tomorrow N.NOM come-NPST C.NML-ACC forget-PST
 ‘Yesterday I forgot that Naraa will come tomorrow.’

- (42) Temporal mismatch in nonfinite complement NC

Öchigdör Bi [*margaash* Naraa ire-h]-iig mart-san.
 yesterday 1SG.NOM tomorrow N.NOM come-INF-ACC forget-PST
 ‘Yesterday I forgot that Naraa is coming tomorrow.’

(43) Temporal mismatch in *-vAl* converbial adjunct

Margaash bid nar [öchigdör Naraa tasalbar av-san bai-val]
 tomorrow 1PL.NOM yesterday N.NOM ticket buy-PFV COP-COND.CVB
 toglolt ruu yav-na.
 concert DIR go-NPST
 ‘If Naraa bought the tickets yesterday, we will be going to the concert tomorrow.’

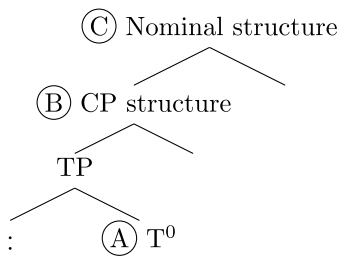
To summarize the adverb placement facts in the four embedded clause types under consideration, each embedded clause type may contain an independent temporal adverb that differs from the tense interpretation in the matrix clause. Further, a theme subject in each embedded clause type may not follow a manner adverb, suggesting that the subject has moved to SpecTP. These two pieces of evidence support the idea that the four embedded clause types contain a TP layer.

Taken together, the scope facts and adverb placement facts indicate that subjects move to SpecTP in Khalkha. I will assume that movement to SpecTP is an instance of A-movement and that this is triggered by an EPP feature on T^0 .

3.2 (Non)finiteness, (non)defective T^0 , and CP structure

The previous subsection has established the presence of TP in all four embedded clause types, thus narrowing the possibilities of variation in their structure to (A) the variety of T^0 (defective vs. nondefective), (B) the amount of CP structure above TP, and (C) the amount of nominal structure above the clause. This is illustrated in (44):

(44) Potential sources of variation in Khalkha embedded clauses



In the remainder of this section, I will investigate (A) and (B), and I will return to (C) in Sect. 4. I will first connect the ability to contain overt tense markers to the presence of nondefective T^0 in finite embedded clauses, and the inability to do so to the presence of defective T^0 in nonfinite embedded clauses. Next, I argue that the presence of the overt complementizer *gej* or *ged(e)g* in finite clauses indicates the presence of a fuller CP structure above TP, while the lack thereof in nonfinite clauses indicates reduced or no structure above TP. This is unsurprising, as nonfinite clauses are commonly argued to be structurally smaller than finite clauses (see, e.g., Wurmbrand 2001, 2024; Adger 2007; McFadden and Sundaresan 2014). Supporting evidence come from i) the distribution of the *wh*-interrogative marker *be/ve* in embedded clauses, and ii) the availability of thematic topic positions in embedded clauses.

Consider first the distribution of overt tense markers. As seen in Sect. 2.3, overt tense markers, such as the nonpast *-nA*, are allowed in finite bare CPs (7) and finite NCs (8) but not nonfinite NCs (9) (examples repeated below).

- (7) Finite bare CP may contain nonpast *-nA*

Bi [Naraa margaash ir-*ne* gej] mart-san.
 1SG.NOM N.NOM tomorrow come-NPST C forget-PST
 ‘I forgot that Naraa will come tomorrow.’

- (8) Finite complement NC may contain nonpast *-nA*

Bi [Naraa margaash ir-*ne* gedg]-iig mart-san.
 1SG.NOM N.NOM tomorrow come-NPST C.NML-ACC forget-PST
 ‘I forgot that Naraa will come tomorrow.’

- (9) Nonfinite complement NC cannot contain nonpast *-nA*

- a. Bi [Naraa-g/giin/Ø margaash ire-*h*]-iig mart-san.
 1SG.NOM N.-ACC/GEN/NOM tomorrow come-INF-ACC forget-PST
 ‘I forgot that Naraa is coming tomorrow.’
 b. *Bi [Naraa-g/giin/Ø margaash ire-*ne*]-g mart-san.
 1SG.NOM N.-ACC/GEN/NOM tomorrow come-NPST-ACC forget-PST
 Int. ‘I forgot that Naraa is coming tomorrow.’

In Sect. 2.3, we also saw that the nonpast tense marker *-nA* cannot appear in converbial adjuncts, for instance (15) (repeated below). Converbial suffixes (e.g., *-hAd* in (15)) thus appear to be in complementary distribution with overt tense markers.

- (15) *-hAd* converbial adjunct cannot contain nonpast *-nA*

Bi [Naraa margaash {ire-*hed* / *ir-*ne*
 1SG.NOM N.NOM tomorrow come-TEMP.CVB come-NPST
 bai-had}] huushuur hii-ne.
 COP-TEMP.CVB khuushuur do-NPST
 ‘I will make khuushuur when Naraa comes tomorrow.’

To reconcile the fact that nonfinite clauses cannot contain an overt tense suffix with the finding in Sect. 3.2 that they contain a TP layer, I propose that the TP layer in nonfinite argument clauses is headed by a phonologically null, defective T^0 . In Khalkha, the defectivity of T^0 is equivalent to the inability of T^0 to be occupied by any of the finite tense markers listed in Table 3.²² I further propose that in converbial adjuncts, defective T^0 is instantiated by the elements that are traditionally called ‘converbs’ in the Mongolic literature (e.g., Janhunee 2012). Under this view, ‘converbs’ are a type of phonologically overt defective T^0 found in nonfinite adjunct clauses. However, I will continue to use the term ‘converb,’ reserving the term ‘overt tense suffix’ only for finite tense suffixes. A partial inventory of tense suffixes in Khalkha is shown in Table 3.

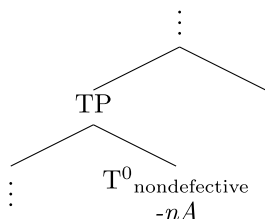
²²That tense is limited in NCs that contain TP but lack CP is not unusual. For instance, Borsley and Kornfilt (2000) report that factive nominalizations in Turkish, which they argue contain TP, only express a nonfuture vs. future tense distinction (see also Kornfilt and Whitman 2011).

Table 3 Selected finite and nonfinite tense markers in Khalkha

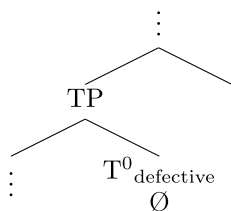
Finite		Nonfinite	
-nA~-Ø	nonpast	-Ø	defective
-IAA	present perfect	-hAd	temporal converb
-sAn~-Ø	simple past	-ngUUt	concomitant converb
-v	past imperfect	-vAl	conditional converb
-jee	past perfect	-tAl	terminative converb

We are now in a position to see how the four types of embedded clauses vary as to the type of T^0 they contain. Finite embedded clauses (e.g., (7) and (8)) contain nondefective T^0 , which expresses a range of tense distinctions and may be phonologically overt, as seen in (45). Nonfinite argument NCs (e.g., (9)) contain a defective T^0 that expresses a limited range of tense interpretations and is always phonologically null, as seen in (46). Converbial adjuncts (e.g., (15)) also contain a defective T^0 , only that it is phonologically overt (i.e., converbs), as seen in (47).

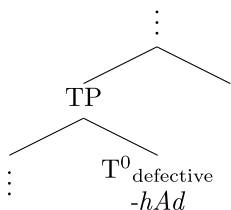
(45) T^0 in finite clauses



(46) T^0 in nonfinite NCs



(47) T^0 in converbial adjuncts

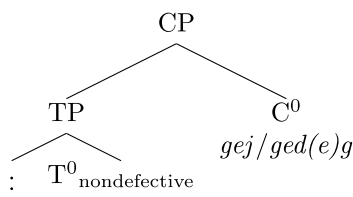


Having established the different types of T^0 contained within each of the four embedded clause types in Khalkha, I turn next to the amount of clausal structure present above TP in them.

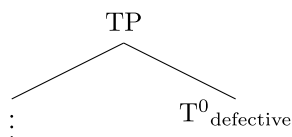
3.3 Clausal structure above TP

There is already evidence that finite clauses contain more structure above TP than nonfinite clauses in Khalkha, due to the simple fact that the former contain the overt complementizer *gej/ged(e)g*, while the latter do not. I show two additional pieces of evidence to support this size distinction between finite and nonfinite clauses in Khalkha. Specifically, I argue that finite clauses contain a full CP structure above TP, while nonfinite clauses lack a full CP structure. This is schematized in (48) and (49).

(48) CP layer in finite clauses



(49) No CP layer in nonfinite clauses



First, I consider whether interrogative markers are permitted in Khalkha embedded clauses, as a means to diagnose the presence of CP structure. This diagnostic is inspired by Saito 2012, who uses the distribution of the interrogative marker *ka* in embedded clauses to investigate the structure of the Japanese clausal periphery. Khalkha is a *wh*-in-situ language, and *wh*-questions contain the *wh*-question marker *be/ve*, as seen in (50).²³

- (50) Chi yuu id-sen be?
 2SG.NOM what eat-PST WHQ
 ‘What did you eat?’

As it is not possible to form an embedded question with a converbial adjunct, I will set them aside in this discussion. (51)–(53) show that complement clauses may serve as embedded questions. However, while finite clauses may optionally contain the *wh*-question marker (51)–(52), nonfinite complement clauses may not (54) cf. (53). For concreteness, I will assume that the overt complementizer *gej/ged(e)g* is in Force⁰, while the *wh*-question marker is in Foc⁰, given that *gej/ged(e)g* follows *be/ve* in the clause.²⁴ See Rizzi 1997 for discussion of different kinds of complementizers in the expanded CP periphery.

²³The form *be* is used after words ending with /m/, /n/ and /w/, while *ve* is used elsewhere.

²⁴Gong 2023a analyzes the *wh*-question marker *be/ve* as instantiating Force⁰, while *gej* occupies another C⁰ above Force⁰. The exact categorial identity of the particle does not affect my main argument that finite embedded clauses contain a fuller CP structure, while nonfinite argument NCs lack CP structure. Note as well that the grammaticality of (54) suggests that it is not the presence of the *wh*-interrogative particle that licenses embedded questions, contrary to Gong 2023a. I leave this issue open for future research.

- (51) Finite bare CP can contain *wh*-question marker *be*

Bi [hen tsuivan id-sen (*be*) geɟ] med-ne.
 1SG.NOM who.NOM tsuivan eat-PST WHQ C know-NPST
 ‘I know who ate tsuivan.’

- (52) Finite NC can contain *wh*-question marker *be*

Bi [hen tsuivan id-sen (*be*) gedɟ]-iig med-ne.
 1SG.NOM who.NOM tsuivan eat-PST WHQ C.NML-ACC know-NPST
 ‘I know who ate tsuivan.’

- (53) Nonfinite argument NC can be embedded questions

Bi [hen tsuivan id-sn]-iig med-ne.
 1SG.NOM who.NOM tsuivan eat-PFV-ACC know-NPST
 ‘I know who ate the tsuivan.’

- (54) Nonfinite argument NC cannot contain *wh*-question marker *be/ve*

*Bi [hen tsuivan id-sen *be/ve*]-g med-ne.
 1SG.NOM who.NOM tsuivan eat-PFV WHQ-ACC know-NPST
 Int.: ‘I know who ate tsuivan.’

The inability of nonfinite complement NCs to contain the *wh*-question marker *be/ve* (and the absence of an overt complementizer *geɟ/ged(e)g* in them) suggests that nonfinite clauses are truncated below CP (i.e., they lack CP structure). Conversely, the ability of finite clauses to contain *be/ve* (and *geɟ/ged(e)g*) suggests that they contain a full CP structure. The size of the clausal periphery is thus a source of difference between finite and nonfinite clauses.

3.4 The availability of thematic topic positions

A last piece of evidence that points towards a size difference between the two classes of embedded clauses is the availability of thematic (or aboutness) topicalization, suggested by Gong (2022, 2023a). Gong (2022, 2023a) shows that a thematic topic position is available in a finite bare CP (55), but not in a nonfinite complement NC (56):²⁵

- (55) Thematic topic reading available in finite bare CP

Bagsh [Bat bol ofis-t=n' hoyor nom-iig unshi-j
 Teacher.NOM B.NOM TOP office-DAT=3.POSS two book-ACC read-CVB
 duusga-san geɟ] hel-sen.
 finish-PST C say-PST

²⁵Gong (2023a) does not discuss what licenses a contrastive topic reading, although she notes that contrastive topics are licensed in a wider range of embedded contexts than thematic topics. Thus, while she argues that thematic topics need to be licensed by raising to a dedicated position within the CP periphery (ThemeP), she assumes that contrastive topics can be licensed in-situ.

✓Contrastive reading: ‘The teacher_i said that Bat (but not other people) finished reading two books in his_i office.’

✓Thematic topic reading: ‘The teacher_i said that as for Bat, he finished reading two books in his_i office.’ (Gong 2022, 199, ex. 288)

(56) Thematic topic reading unavailable in nonfinite argument NC

Bagsh [Bat bol tavan nom unshi-h]-iig hel-sen.
Teacher.NOM B.NOM TOP five book read-INF-ACC say-PST

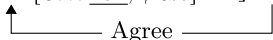
✓Contrastive reading: ‘The teacher said that Bat (but not other people) read 5 books.’

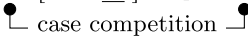
✗Thematic topic reading: ‘The teacher said that as for Bat, he read five books.’ (Gong 2022, 199–200, ex. 289)

Assuming that a thematic topic position is located in the expanded CP periphery, for instance TopicP (Rizzi 1997) or ThemeP (Gong 2023a), the asymmetry between (55) and (56) can be interpreted to suggest that finite clauses containing the overt complementizer *gej* are full CPs, while nonfinite clauses lacking *gej* lack CP structure. Gong does not discuss thematic topicalization in converbial adjuncts and finite NCs. Based on her analysis, one would predict that the behavior of thematic topicalization in finite NCs should parallel that of finite bare CPs, since both clause types contain an overt complementizer *gej/ged(e)g*, while thematic topicalization in converbial adjuncts should parallel that in nonfinite NCs, since both clause types lack *gej/ged(e)g*.

3.5 On nominative case assignment

Before I end this section, I will briefly discuss the significance of the analysis above for understanding the relationship between finiteness and nominative case assignment in Khalkha. There are two major theories as to how nominative case is assigned in a language. The functional-head licensing approach, schematized in (57), takes nominative case to be assigned to a nominal by finite T^0 via an Agree relation (e.g., George and Kornfilt 1981; Chomsky 1980, 2000, 2001). Under this account, T^0 bears a nominative Case feature and an unvalued ϕ -feature, while the subject DP bears ϕ -features and an unvalued Case feature (57a). To value its unvalued ϕ -feature, T^0 searches its c-command domain for the closest active goal—this will typically be the subject DP that is first merged in SpecVoiceP. T^0 Agrees with the subject DP, as shown in (57b), resulting in the subject DP’s ϕ -features being copied onto T^0 , and the unvalued Case feature of the subject DP being valued by T^0 . Alternatively, under a configurational view of case (Yip et al. 1987; Marantz 1991; Baker 2015), nominative case is assigned to a nominal within the TP domain that has gone through the derivation without being assigned any case; this is known as unmarked case. (58) sketches how configurational case assignment takes place in a root clause, following Baker and Vinokurova 2010. Upon spell-out of the CP phase, accusative case is assigned as a dependent case to the lower of two case-unmarked DPs (typically the direct object) in a local asymmetric c-command relation (58a), after which nominative case is assigned as an unmarked case to any DP (typically the subject) that has not been assigned any case (58b).

- (57) Functional-head nominative case assignment
- T^0 bears unvalued ϕ ; subject DP bears unvalued Case
 $[TP \ [vP \ DP_{[Case: _, \phi:3SG]} \dots] \ T^0_{[Case:NOM, \phi: _]}]$
 - T^0 Agrees with subject DP and assigns it nominative case
 $[TP \ [vP \ DP_{[Case:NOM, \phi:3SG]} \dots] \ T^0_{[Case:NOM, \phi:3SG]}]$


- (58) Configurational accusative case assignment
- Accusative case assigned to DP_1 under case competition with DP_2
 $[TP \ DP_2_{[Case: _]} \dots [vP \ DP_1_{[Case: ACC]} \dots]]$

 - Nominative case assigned to DP_2 as an unmarked case
 $[TP \ DP_2_{[Case: NOM]} \dots [vP \ DP_1_{[Case: ACC]} \dots]]$

If nominative case assignment is tied to the presence of finite, nondefective T^0 , then the wide distribution of nominative embedded subjects in Khalkha is surprising.²⁶

Although Khalkha lacks overt subject-verb agreement, the defectivity of T^0 in nonfinite clauses can be inferred from their inability to express overt tense. Nonetheless, nonfinite clauses, such as nonfinite complement NCs and converbial adjuncts, may still contain nominative unmarked subjects. This suggests that nominative case assignment is independent of whether T^0 is defective or nondefective. Instead, the Khalkha data supports a configurational account in which nominative is assigned as an unmarked case within TP. For further discussion on the dissociation between finite T^0 and nominative case assignment in Khalkha and in Kalmyk Oirat—a related Mongolic language with overt subject-verb agreement—see Lim (2024b).

3.6 Summary

In this section, I have argued that all four embedded clause types in Khalkha contain TP, thereby narrowing the space of variation in clausal structure to the type of T^0 and the presence or absence of CP structure contained in embedded clauses. I have provided additional arguments for the finite vs. nonfinite distinction in Khalkha embedded clauses discussed in Sect. 2: finite clauses contain a nondefective T^0 and a full CP structure, while nonfinite clauses contain a defective T^0 and lack any CP structure. Table 4 summarizes the findings of this section.

The clause structure of finite clauses (including finite bare CPs and finite NCs), nonfinite NCs, and converbial adjuncts is shown in (59), (60), and (61), respectively.

²⁶ A reviewer raises the intriguing possibility that Khalkha may exhibit multiple types of “nominative” subjects, each with distinct syntactic and semantic properties. In Turkish, for instance, “nominative” subjects in root clauses and in embedded NCs differ in their morphosyntactic features and semantics/pragmatics, suggesting the existence of multiple “nominative” cases (e.g., Kornfilt 2020). This view opens up the possibility of maintaining an Agree-based approach to nominative in Khalkha where one type of nominative is assigned by nondefective T^0 and another type of nominative is a default case that co-occurs with defective T^0 . While a systematic investigation of this hypothesis lies beyond the scope of this paper, I thank the reviewer for helpful discussion of this idea.

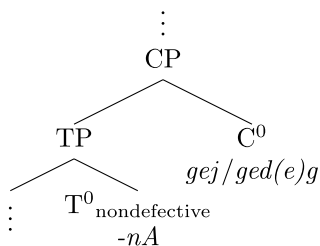
Table 4 Clausal properties of Khalkha embedded clauses

Correlates of finiteness	Clause type			
	Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
Contains overt C^0 <i>gej/ged(e)g</i>	✓	✓	✗	✗
Able to contain overt tense suffixes	✓	✓	✗	✗
Able to contain WHQ <i>belve</i>	✓	✓	✗	<u>-1</u>
Able to contain thematic topics	✓	-	✗	<u>-2</u>

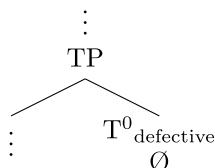
¹This dash represents the fact that embedded question formation is not possible with converbial adjuncts.

²The dashes in this row represent the two clause types for which Gong (2022, 2023b) does not provide data regarding thematic topicalization.

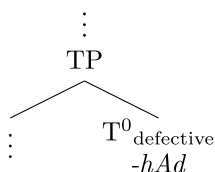
(59) Clausal structure in finite clauses



(60) Clausal structure in nonfinite NCs



(61) Clausal structure in converbial adjuncts



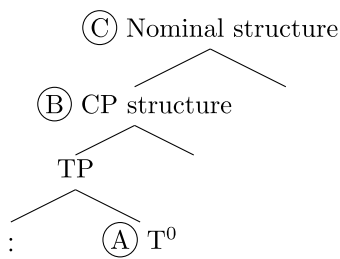
In the next section, I will probe further into the nominal layers present or absent in the four embedded clause types in Khalkha.

4 Nominal superstructure in embedded clauses

Up to this point, I have discussed how the four embedded clause types in Khalkha may differ as to (A) the type of T^0 they contain, and (B) the amount of CP structure they contain (see (44), repeated below). In this section, I further argue that argument

NCs and converbial adjunct NCs actually differ in ③ the amount of nominal structure they contain.

(44) Potential sources of variation in Khalkha embedded clauses



As I will show below, while finite NCs, nonfinite NCs, and converbial adjuncts may host possessive enclitics, only finite and nonfinite NCs may additionally bear structural case marking and contain genitive subjects. Following works such as Alexiadou 2001, 2020, Alexiadou et al. 2010 and Baker 2011, I will take the presence or absence of a nominal property to reflect the presence or absence of a dedicated functional layer. It follows that, while both clause types are nominalized, argument NCs contain more nominal structure than converbial adjunct NCs.

My analysis is not without precedent; while the exact details differ, Kornfilt 2006 also argues that adjunct NCs in Turkish lack some nominal structure found in argument NCs. My analysis is also supported by parallel observations from the literature on so-called Small Nominals, where it is also proposed that variation in nominal properties reflects variation in the amount of nominal functional structure present (Pereltsvaig 2006, et seq.) I propose that, just as nominals may vary as to whether they are full DPs or small nominals that lack DP (Pereltsvaig and Lyutikova 2014; Lyutikova and Pereltsvaig 2015), nominalized clauses may also vary as to whether they are full DPs or small nominalized clauses (SNCs) which lack DP. Specifically, I propose that argument NCs are full DPs that contain an intermediate nominal layer PossP and nominalizing nP, while converbial adjuncts are SNCs—smaller PossPs that contain nominalizing nP but lack DP. This is schematized in (62).

(62) Nominal structure of embedded clauses in Khalkha

- a. Argument NCs (finite NC, nonfinite NC)
DP > PossP > nP > clausal structure
- b. Converbial adjuncts
PossP > nP > clausal structure

I begin with some background on the analysis of small nominals, before discussing how the insights from research on small nominals may be applied to understanding nominal structure in Khalkha NCs. I end the section with a short discussion of how the findings are relevant for understanding genitive case assignment in Khalkha.

4.1 Small nominals

The idea that a single language may contain full DPs and smaller-sized nominals is familiar from the literature on noun incorporation and pseudo-noun incorporation

(for recent overviews, see Massam 2009, 2017; Johns 2017). However, a different version of this idea comes from Pereltsvaig (2006), who argues that standalone (non-pseudo-incorporated) nominals may also vary as to whether they are full DPs or small nominals—nominals that lack a DP layer. Pereltsvaig argues that not all small nominals are incorporated or pseudo-incorporated nouns; for instance, small nominals need not be adjacent to the verb and can be displaced by certain operations that target nonreferential nominals (i.e., nominals which lack a DP layer). For example, an operation known as Approximative Inversion in Russian may only target small nominals, for instance *čelovek pjat* ‘five people’ in (63). Pereltsvaig argues that the ill-formedness of (63) with the referential pronoun *ix* ‘them’ lends support to the idea that the topicalized argument is nonreferential, and hence a small nominal.

(63) Russian

Čelovek pjat, my eščë ne videli (*ix).

people five we still not saw them

‘As for a group of approximately five people, we haven’t seen such a group yet.’
(Pereltsvaig 2006, 474, fn.32)

The view that a single language may contain both full DPs and SNs stands in opposition to two influential views of nominal structure: that argument nominals in all languages must be DPs, following Longobardi (1994); or that languages can be classified according to whether they are a DP-language or NP-language, following Chierchia (1998) and Bošković (2008).²⁷

Another language that has been argued to contain both full DPs and small nominals is Tatar (Turkic), where the variation can be observed in the possessive domain (Pereltsvaig and Lyutikova 2014; Lyutikova and Pereltsvaig 2015). In the so-called *ezafe-3* possessive construction in (64a), the possessor is marked with genitive case and possessor agreement is reflected on the head noun. (64a) denotes an apple that belongs to children. In the *ezafe-2* possessive construction in (64b), morphology that is homophonous with the third singular possessive agreement suffix (-sı) is also present on the head noun, but the possessor does not receive any case marking. (64b) denotes an apple that is associated with children.

(64) Tatar

a. *Ezafe-3* possessive nominal

bala-lar-nıŋ alma-sı

child-PL-GEN apple-3.POSS

‘(the) children’s apple’

b. *Ezafe-2* possessive nominal

bala-lar alma-sı

child-PL apple-3.POSS

‘children’s apple (i.e. an apple for children)’

(Lyutikova and Pereltsvaig 2015, 300, ex. 20)

²⁷Even within such an approach, it is sometimes murky how to classify individual languages, such as Turkish. For example, Bošković and Şener (2014) argue that Turkish should be classified as an NP-language, while their view has been criticized by Kornfilt (2018a,b).

Further, Pereltsvaig and Lyutikova show that while *ezafe-3* objects are obligatorily accusative-marked (65a), *ezafe-2* objects may be case-unmarked (65b) or accusative-marked (65c).

(65) Tatar

- a. *Ezafe-3* object with obligatory accusative marking
 Min [bala-lar-niŋ tabib-i*(-n)] ezl-i-m.
 I child-PL-GEN doctor-3.POSS-ACC look.for-PRES-1SG
 ‘I am looking for a/the children’s doctor.’
- b. *Ezafe-2* object without accusative marking
 Min [bala-lar tabib-i] ezl-i-m.
 I child-PL doctor-3.POSS look.for-PRES-1SG
 ‘I am looking for a pediatrician.’
- c. *Ezafe-2* object with accusative marking
 Min [bala-lar tabib-i-n] ezl-i-m.
 I child-PL doctor-3.POSS-ACC look.for-PRES-1SG
 ‘I am looking for {a certain/the} pediatrician.’

(Lyutikova and Pereltsvaig 2015, 316, ex. 60)

Finally, Pereltsvaig and Lyutikova (2014, 206) assume that case markers are found on K^0 , which selects only for DPs, and that DPs in Tatar are subject to the Case Filter.

For *ezafe-3* constructions, Pereltsvaig and Lyutikova propose that they are full DPs which contain full DP possessors. Since the *ezafe-3* construction is a full DP, it is subject to the Case Filter and is obligatorily accusative marked. Since the *ezafe-3* possessor is also a full DP, it is also subject to the Case Filter and is obligatorily genitive marked. The authors argue that the source of genitive case is D^0 of the *ezafe-3* construction, which Agrees with the DP possessor, raises it to SpecDP,²⁸ and assigns it genitive case. They treat the *ezafe-3* marker on the head noun (e.g., *-sı* in (64a)) as ϕ -agreement on D^0 .

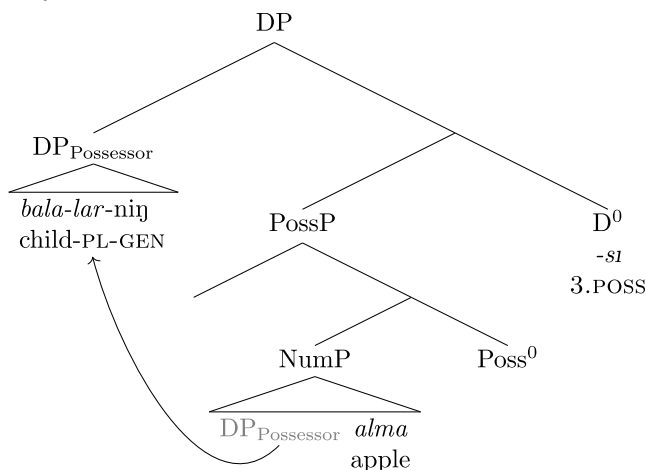
For *ezafe-2* constructions, Pereltsvaig and Lyutikova propose that these may either be PossPs lacking a DP layer (i.e., small nominals) or PossPs embedded under a null DP. They also propose that *ezafe-2* possessors are PossPs lacking a DP layer. They analyze case-unmarked *ezafe-2* objects as bare PossPs (e.g., (65b)), and accusative-marked *ezafe-2* objects as full DPs (e.g., (65c)). Although the authors are not entirely explicit on this point, they seem to suggest that even when PossPs are selected by a null D^0 , this head does not Agree with the possessor and does not assign genitive case. Since *ezafe-2* possessors cannot be assigned any Case—given the lack of (genitive-assigning) D^0 —they cannot be full DPs due to the Case Filter. As support for this view, they note that *ezafe-2* possessors cannot be pronouns or proper names, which are full DPs and are subject to the Case Filter. Thus, *ezafe-2* possessors must themselves be small nominals (i.e., PossPs). The authors propose that in *ezafe-2*, the

²⁸Pereltsvaig and Lyutikova (2014) suggest two possible positions where the *ezafe-3* possessor is first merged: in SpecNP, which corresponds to an Agent/Creator/Owner interpretation of the possessor, or as a complement to N^0 , which corresponds to an internal argument interpretation of the possessor. For *ezafe-2* possessors, the authors argue that they are first merged in SpecPossP.

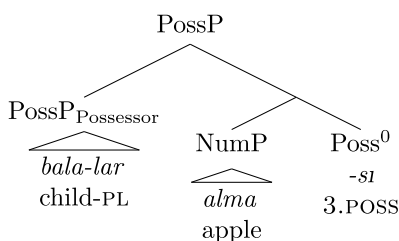
possessor occupies SpecPossP, and that the suffix on the head noun (e.g., *-sɪ* in (64b) and (65b)–(65c)) is an *ezafe-2* marker realized on Poss⁰.²⁹ (66a)–(66c) illustrate the structures of the objects in (65), adapted from Pereltsvaig and Lyutikova 2014.

(66) Three possessive structures in Tatar

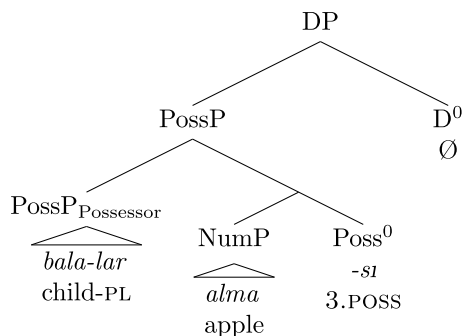
a. *Ezafe-3 DP*



b. *Ezafe-2 PossP*



c. *Null DP merged above ezafe-2*



²⁹The authors gloss the *ezafe-2* as 3rd person possessive agreement, but are agnostic as to whether the *ezafe-2* marker represents true ϕ -agreement with the third person possessor argument. As *ezafe-2* constructions cannot contain first and second person possessor arguments, it is not possible to test for covarying agreement.

That the possessors are located in different structural positions is supported by evidence such as adjective placement, as seen in (67). In (67a), the *ezafe*-3 possessor, marked with genitive case, obligatorily precedes the adjective *kük* ‘blue,’ while in (67b) the *ezafe*-2 case-unmarked possessor obligatorily follows the same adjective. While the authors do not specify the position of the adjective, they assume that it occupies the same position in (67a) and (67b). The difference in the distribution of the adjective thus suggests that the genitive-marked possessor is structurally higher (e.g., in SpecDP) than the case-unmarked one (e.g., located in SpecPossP).

(67) Tatar

- a. *Ezafe*-3 possessor obligatorily precedes adjective
 (*kük) bala-lar-niŋ (kük) itek-lär-e
 blue child-PL-GEN blue boot-PL-3.POSS
 ‘(the) children’s blue boots’
- b. *Ezafe*-2 possessor obligatorily follows adjective
 (kük) bala-lar (*kük) itek-lär-e
 blue child-PL blue boot-PL-3.POSS
 ‘blue children’s boots’

(Lyutikova and Pereltsvaig 2015, 300–301, ex. 21)

In sum, possessive nominals in Tatar come in two sizes: full DPs (with Agreeing D⁰ or null, non-Agreeing D⁰) which contain PossPs, and reduced PossPs which lack DP. In the next subsection, I show how Pereltsvaig and Lyutikova’s analysis of small nominals in Tatar can be applied to understand a similar structural variation in Khalkha NCs.

4.2 Nominal structure in embedded clauses

As discussed in Sect. 2.2, Khalkha possessive nominals may contain a genitive-marked pronoun (6a), a ϕ -agreeing possessive enclitic (6b), or both (6c) (examples repeated below). Notice that as complements of verbs, possessive nominals are obligatorily assigned differential accusative case, arguably because they are full DPs and interpreted as specific and definite.

(6) Three ways to express possession in Khalkha

- a. Genitive pronoun
 Chi *minii* nom-*iig* unsh-san.
 2SG.NOM 1SG.GEN book-ACC read-PST
- b. Possessive enclitic
 Chi nom-*iig*=*min*’ unsh-san.
 2SG.NOM book-ACC=1SG.POSS read-PST
- c. Genitive pronoun + possessive enclitic
 Chi *minii* nom-*iig*=*min*’ unsh-san.
 2SG.NOM 1SG.GEN book-ACC=1SG.POSS read-PST
 ‘You read my book.’

To the best of my knowledge, Khalkha does not appear to have small nominals resembling *ezafe*-2 possessives in Tatar. Barrie and Kang 2022, Kang 2023, and Barrie 2024 discuss prosodic evidence for both full DPs and smaller nPs in Khalkha,³⁰ but they analyze reduced nPs as pseudo-incorporated nominals rather than standalone small nominals as defined by Pereltsvaig 2006. Pending further investigation, this paper assumes the absence of non-pseudo-incorporated small nominals in Khalkha's purely nominal domain. However, I argue that the full DP-small nominal distinction emerges in the domain of nominalized clauses. To do this, I will examine how the four embedded clause types in Khalkha pattern with respect to the three nominal properties found on possessive nominals: i) the availability of possessive enclitics, ii) the availability of genitive case assignment, and iii) the ability to bear structural case marking. Previewing my findings, I show that finite CPs do not exhibit any of the three abovementioned nominal properties, complement NCs display all three nominal properties, and converbial adjuncts display only one of the three nominal properties—namely, the ability the host possessive enclitics. Following Borsley and Kornfilt 2000; Alexiadou 2001, 2020; Alexiadou et al. 2007, 2010; Kornfilt and Whitman 2011, a.o., I adopt the idea that the clausal properties of NCs stem from clausal functional structure (e.g., vP, VoiceP, AspP, TP, CP) below the level of nominalization, while their nominal properties arise from nominal functional structure (e.g., nP, PossP, DP) above the clausal level. Nominalization is mediated by a categorizing (Marantz 2001) or “category-switching” (Rouveret 2023) head, n^0 , projected directly above the clausal level, which serves to nominalize the clause. In my analysis, n^0 is null in nonfinite NCs but triggers allomorphy of C^0 (*ged(e)g*) in finite NCs (see (10)).

I take the variation in nominal properties across different types of NCs to reflect differences in the amount of nominal functional structure above n^0 . This analysis argues against a direct DP-shell approach to Khalkha NCs, which posits DP as the sole nominal layer above the clause (e.g., Borsley and Kornfilt 2000; Alexiadou et al. 2010; Ótót-Kovács 2023). Such an approach would be ill-equipped to account for the variation in nominal properties across NC types without permitting differences in the extent of nominal functional structure.

Considering each embedded clause type in turn, first, finite bare CPs do not exhibit any of the three nominal properties of possessive nominals (68). I take this to indicate the lack of any nominal structure in them. Thus, I will exclude finite bare CPs from the following discussion to focus on the other three embedded clause types in Khalkha.

- (68) Finite bare CP may not contain a genitive subject, bear case marking, or host a possessive enclitic

Bi [Naraa-g/*giin unta-j bai-san gej](*-iig)(*=n')
 1SG.NOM N.-ACC/GEN sleep-PROG COP-PST C-ACC=3.POSS
 med-ne.
 know-NPST
 ‘I know that Naraa was sleeping.’

³⁰Their analysis focuses on simplex nominals. They cite Guntsetseg (2016), who observes that pseudo-incorporated nouns in Khalkha lack demonstratives, possessors, postpositions, case markers, and plural marking. See also Driemel 2020, 2023 on pseudo-noun incorporation in Khalkha.

Finite (69) and nonfinite (70) NCs fully parallel possessive nominals in exhibiting all three nominal properties: they may host possessive enclitics, contain a genitive subject, and bear structural case marking. Thus, I take both finite and nonfinite NCs to be full DPs, like possessive nominals.

- (69) Finite NC with genitive subject, accusative case and possessive enclitic

Naraa *minii* unta-j bai-san gedg]-iig(=*min*)
 N.NOM 1SG.GEN sleep-PROG COP-PST C.NML-ACC=1SG.POSS
 med-ne.
 know-NPST
 ‘Naraa knows that I was sleeping.’

- (70) Nonfinite NC with genitive subject, accusative case and possessive enclitic

Naraa [*minii* unta-j bai-sn]-iig(=*min*) med-ne.
 N.NOM 1SG.GEN sleep-PROG COP-PFV-ACC=1SG.POSS know-NPST
 ‘Naraa knows that I was sleeping.’

Strikingly, converbial adjuncts exhibit only a subset of these properties. First, while converbial adjuncts may host possessive enclitics, they may not contain genitive subjects. This is seen in (71).³¹

- (71) Converbial adjunct may host possessive enclitic but cannot contain genitive subject

Chi [namaig/**minii* hicheel-Ø=ee hii-j
 2SG.NOM 1SG.ACC/1SG.GEN lesson-ACC=RP do-PROG
 bai-had](=*min*) unt-san.
 COP-TEMP.CVB=1SG.POSS sleep-PST
 ‘When I was studying (lit. doing my lesson), you fell asleep.’

Additionally, converbial adjuncts cannot be marked with any case marking, such as accusative case with a matrix transitive verb in (72), or dative case with a matrix psych verb in (73).³²

³¹I represent accusative subjects in converbial adjuncts with possessive clitic doubling due to the fact that my consultants disprefer clitic doubling with nominative subjects (see footnote 13), and that genitive subjects are disallowed in them.

³²One might argue that converbial adjuncts are not overtly case-marked because they are inherently case-marked. Umetani (2019) proposes a decompositional analysis for certain converbs, particularly those with transparent morphology. For instance, he analyses the temporal converb *-hAd* as consisting of the infinitive/futuristic aspect marker *-h* and the dative case suffix *-Ad*. If this analysis were correct, one might predict that dative-marked adjuncts could function as complements of psych predicates like *uurla-* ‘be angry,’ which take dative-marked arguments.

However, the example in (i), though grammatically acceptable, fails to convey the intended reading where the embedded subject *Naraa*’s sleeping directly causes the first-person subject’s anger. Instead, consultants report that the source of anger in (i) is unclear or attributed to a *pro*-dropped argument. This suggests that the converb does not contain an active case marker. While I leave open the possibility that some converbs may include fossilized case markers due to their etymology (see footnote 15), I do not assume that converbial adjuncts are inherently case-marked.

(72) Converbial adjunct cannot be marked with accusative case

- a. *Bi [Naraa-g hicheel-Ø=ee hii-j
 1SG.NOM N.-ACC lesson-ACC=RP do-PROG
 bai-had]-iig(=n') med-ne.
 COP-TEMP.CVB-ACC=3.POSS know-NPST
 Int. 'I know when Naraa was studying.'
- b. Bi [Naraa-g huushuur ide-j bai-had](=*-iig)(=n')
 1SG.NOM N.-ACC khuushuur eat-PROG COP-TEMP.CVB-ACC=3.POSS
 nom unshi-j bai-san.
 book read-PROG COP-PST
 'I was reading a book while Naraa was eating khuushuur.'

(73) Converbial adjunct cannot be marked with dative case

- a. *Bi [Naraa-g unta-j bai-had]-ad(=n')
 1SG.NOM N.-ACC sleep-PROG COP-TEMP.CVB-DAT=3.POSS
 uurla-san.
 be.angry-NPST
 Int. 'I was angry at (the time when) Naraa (was) sleeping.'
- b. Bi [Naraa-g unta-j bai-had](=*-ad)(=n') Dorj-id
 1SG.NOM N.-ACC sleep-PROG COP-TEMP.CVB-DAT=3.POSS D.-DAT
 uurla-san.
 be.angry-NPST
 'I was angry at Dorj when Naraa was sleeping.'

The ability of converbial adjuncts to host possessive suffixes indicates the presence of some nominal structure. However, they lack the additional nominal layers found in finite and nonfinite complement NCs that enable genitive case assignment and allow the clause itself to be case-marked. This implies that converbial adjuncts are structurally smaller in the nominal domain compared to finite and nonfinite complement NCs.

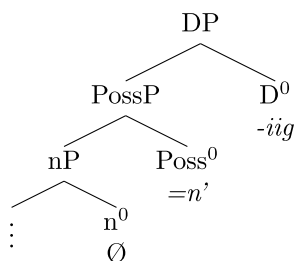
4.3 Two types of nominal structure

To summarize, finite and nonfinite NCs exhibit the full range of nominal properties found in possessive nominals, while converbial adjuncts exhibit only a subset—namely, the ability to host possessive enclitics. I propose that this variation arises because finite and nonfinite NCs are full DPs that contain PossP and nP (74), whereas converbial adjuncts are reduced PossPs that contain nP but lack DP (75) (structures repeated below). Note that I assume n^0 to be phonologically null in Khalkha NCs, possessive enclitics (e.g., third possessive =*n*') to occupy Poss⁰ as a result of Agree

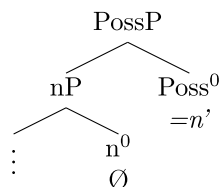
-
- (i) Bi [Naraa-g unta-j bai-had](=n') uurla-san.
 1SG.NOM N.-ACC sleep-PROG COP-TEMP.CVB=3.POSS be.angry-NPST
 'I was angry (at something) when Naraa was sleeping.'
 Does not mean: 'I was angry at Naraa sleeping.'

with the embedded subject, and case suffixes (e.g., accusative *-iig*) to occupy D^0 . Additionally, I assume that a postsyntactic reordering operation also applies to NCs to derive the POSS-CASE order, although this reordering is not depicted in (74) and (75). Under this analysis, I propose to call converbial adjuncts small nominalized clauses (SNCs), under similarity with small nominals (Pereltsvaig 2006, et seq.).

(74) Nominal structure in finite/nonfinite NCs



(75) Nominal structure in converbial adjuncts



4.4 On genitive case assignment

Before concluding this section, I briefly highlight how the analysis of nominal structure impacts our understanding of genitive case assignment in Khalkha. Recall that the subject of a finite or nonfinite argument NC may receive genitive case, while the subject of a converbial adjunct NC may not. I have argued that argument NCs contain a full DP layer, whereas converbial adjunct NCs are reduced PossPs lacking a DP layer. This suggests that the DP layer is crucial for genitive case assignment in the nominal domain, such as to possessor arguments.

This observation aligns with two prominent theories of genitive case assignment: (i) Agree-based, functional head theories, where genitive case is assigned via an Agree relation with a functional head (e.g., Asarina 2011; Baker and Vinokurova 2010; Bošković 2008; Hale 2002; Kornfilt 2003; Laszakovits 2018; Ótrott-Kovács 2020, a.o.), and (ii) configurational case theories, where genitive case is treated as an unmarked case within the DP phase (e.g., Baker 2015; Levin and Preminger 2015; Marantz 2001). These two approaches are schematized in (76) and (77).

(76) Functional head genitive case assignment

- a. D^0_2 bears an unvalued ϕ -feature; DP_1 bears an unvalued Case feature
 $[DP_2 [PossP DP_1[Case: _ , \phi: 3SG] \dots] D^0_2[Case: GEN, \phi: _]]$

Table 5 Nominal properties of Khalkha embedded clauses

Correlates of nominal structure	Clause type			
	Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
Able to host possessive enclitics	✗	✓	✓	✓
Able to contain genitive subject	✗	✓	✓	✗
Able to be assigned case	✗	✓	✓	✗

- b. D^0_2 Agrees with DP_1 and assigns it genitive case

D^0_2 Agrees with DP_1 and assigns it genitive case

$[DP_2 [PossP DP_1[Case:GEN, \phi:3SG] \dots] D^0_2[Case:GEN, \phi:3SG]]$
 $\quad \quad \quad \uparrow \quad \quad \quad \text{Agree} \quad \quad \quad \downarrow$

(77) Configurational genitive case assignment

- a. No dependent case assigned

$[DP_2 \dots DP_1[Case: _]]$

- b. Genitive case assigned as an unmarked case within DP

$[DP_2 \dots DP_1[Case:GEN]]$

For Khalkha, adopting an Agree-based view of genitive case assignment (76) would require one to conclude that the Agree operation responsible for genitive case is distinct from the one that yields possessor agreement morphology. Evidence comes from converbial adjuncts (70), which exhibit a double dissociation between genitive case assignment and possessor agreement (realized as possessive clitics). First, non-genitive subjects, such as accusative subjects, are compatible with possessive clitics in converbial adjuncts (and argument NCs). Second, genitive case cannot be assigned to subjects in converbial adjuncts despite the presence of possessive clitics. This dissociation aligns more naturally with a configurational approach to genitive case (77), where genitive is assigned as an unmarked case in the DP domain to DPs that have not been assigned any case. This explains why environments with possessor agreement do not necessarily permit genitive case assignment, as the two phenomena stem from distinct mechanisms. For further discussion on the dissociation of genitive case assignment from possessor agreement morphology, see Lim (2025).

4.5 Summary and discussion

This section has demonstrated that while finite and nonfinite complement NCs exhibit the full range of nominal properties shared with possessive nominals, converbial adjuncts exhibit only a subset of these properties. Following the view that nominal properties in nominalizations correspond to the presence of nominal structure (Korfilft and Whitman 2011), I propose that complement NCs are full DPs containing PossP, whereas converbial adjuncts are reduced PossPs lacking a DP layer. Table 5 summarizes these findings.

Interestingly, the DP/PossP distinction aligns with the argument/adjunct distinction in Khalkha. While Khalkha arguments can be full DPs or pseudo-incorporated

Table 6 Morphosyntactic properties of Khalkha embedded clauses

Correlates of...	Clause type			
	Finite bare CP	Finite NC	Nonfinite NC	Converbial adjunct
Finiteness				
Contains overt C ⁰ <i>gej/ged(e)g</i>	✓	✓	✗	✗
Able to contain overt tense suffixes	✓	✓	✗	✗
Able to contain WHQ <i>belve</i>	✓	✓	✗	-
Able to contain thematic topics	✓	-	✗	-
Nominal structure				
Able to host possessive enclitics	✗	✓	✓	✓
Able to contain genitive subject	✗	✓	✓	✗
Able to be assigned case	✗	✓	✓	✗

nouns (Guntsetseg 2016; Driemel 2020, 2023; Barrie and Kang 2022; Kang 2023; Barrie 2024), it remains unclear whether standalone small nominals can serve as arguments in Khalkha. As Pereltsvaig and Lyutikova (2014) argue for Tatar, both DPs and PossPs may serve as arguments in the language, see, for instance, (65). In the next two sections, I offer some predictions as to where one may find small nominals or SNCs in a given language. Before doing so, in the next section, I provide sketches of the structures of the four embedded clause types under investigation.

5 Mixed extended projections in Khalkha

In this section, I synthesize the findings from the previous two sections to propose a micro-typology of nominalized clauses in Khalkha. I also address a striking gap in the data: the absence of a nominalization in which a full CP structure is embedded under a reduced PossP structure. Building on Svenonius (2021), I argue that this gap can be explained by a broader constraint on embedding proposed by Williams (2003), which has also been invoked to account for other size-based asymmetries (Keine 2020; Poole 2023).

5.1 Putting clausal and nominal structure together

In Sect. 3 and Sect. 4, I have shown how four types of embedded clauses in Khalkha vary in the amount of clausal and nominal structure they contain. The findings are summarized in Table 6, and each clause type is illustrated again in (78)–(81).

- (78) Bi [Naraa margaash ir-ne gej] hel-sen.
 1SG.NOM N.NOM tomorrow come-NPST C say-PST
 ‘I said that Naraa will come tomorrow.’

- (79) Bi [Naraa-giin margaash ir-ne gedg]-iig(=n')
 1SG.NOM N.-GEN tomorrow come-NPST C.NML ACC=3.POSS
 mart-san.
 forget-PST
 'I forgot that Naraa will come tomorrow.'
- (80) Bi [Naraa-giin margaash ire-h]-iig(=n') mart-san.
 1SG.NOM N.-GEN tomorrow come-INF -ACC=3.POSS forget-PST
 'I forgot that Naraa is coming tomorrow.'
- (81) Bi [ter hicheel-Ø=ee hii-j bai-had](=n')
 1SG.NOM 3SG.NOM lesson-ACC=RP do-PROG COP-TEMP.CVB =3.POSS
 unt-san.
 sleep-PST
 'When she/he was studying, I fell asleep.'

I propose that converbial adjuncts adjoin to vP—a position below VoiceP where the external argument is introduced.³³ This is supported by two observations. First, as shown in (82), a converbial adjunct cannot follow a manner adverb like *hurda* 'quickly,' which adjoins to vP. However, in (83), a converbial adjunct may precede or follow an aspectual adverb like *dandaa* 'always,' which adjoins to AspP.

- (82) Bat { *hurda } [eej-iig=ee hair-had] { hurda }
 B.NOM quickly mother-ACC=RP return-TEMP.CVB quickly
 ger-iin daagavr-Ø=aa hii-sen.
 home-GEN assignment-ACC=RP do-PST
 'When his_j mother returned (home), Bat_j quickly did his homework.'
- (83) Misheel { dandaa } [eej-iig=ee har'-tal] { dandaa } video
 M. always mother-ACC=RP return-TERM.CVB always video
 togloom toglo-dog.
 game play-HAB
 'Misheel_j always plays video games until her_j mother returns (home).'

The contrast between (82) and (83) suggests that converbial adjuncts adjoin to a position at least as high as vP.³⁴ Further, (82) and (83) also show that the possessor of the embedded subject within the converbial adjunct can be bound by the matrix subject via reflexive-possessive marking. Assuming that subject-oriented anaphoric binding in Khalkha requires local c-command by a subject located in SpecVoiceP

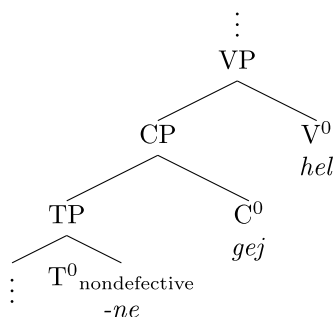
³³In contrast, Peters (2024, 19) proposes that converbial adjuncts adjoin to a position above VoiceP but below TP. Under this analysis, converbial adjunct subjects are never locally c-commanded by a matrix subject argument in SpecVoiceP, and the fact that converbial adjunct subjects and possessors located within such subjects may be anaphorically bound by matrix subjects would require an alternative explanation than that proposed by Gong 2023b; Gong and Despić 2024 (see below for some discussion).

³⁴Given that both *hurda* 'quickly' and converbial adjuncts adjoin to vP, in principle, we predict that the positions of these constituents would be interchangeable. However, *hurda* may not precede the converbial adjunct, as shown in (82). I take this to imply that other factors may be involved in determining the order of manner adverbs and converbial adjuncts.

(Gong 2023b; Gong and Despić 2024), this suggests that converbial adjuncts adjoin below VoiceP. Taken together, I assume that converbial adjuncts uniformly adjoin to vP, and that they may scramble higher within the main clause, for instance, in (83), where the converbial adjunct precedes the aspectual adverb *dandaa* ‘always’.

In (84)–(87), I provide partial structures for each embedded clause type, illustrating only the TP, CP, and nominal layers within them. For argument clauses, I include the matrix predicate which selects for them, and for converbial adjuncts, I include the matrix AspP projection to which they are adjoined. A finite bare CP contains a full CP with nondefective T⁰, but lacks any nominal structure at all, as seen in (84). A finite NC contains a full CP with nondefective T⁰, and is nominalized by a full DP structure (85). A nonfinite NC contains no CP structure and defective T⁰, but is again nominalized by a full DP (86). Lastly, a converbial adjunct also contains no CP structure, but contains an overt defective T⁰ (i.e., a converb) and is nominalized by a reduced PossP structure (87). Converbial adjuncts are thus SNCs.

(84) Finite bare CP (78)



(85) Finite NC (79)

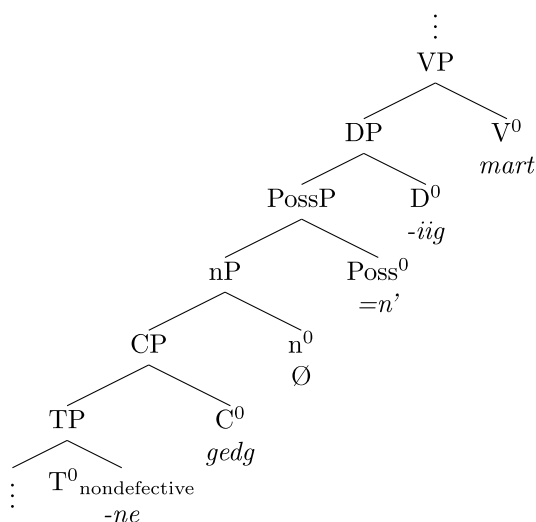
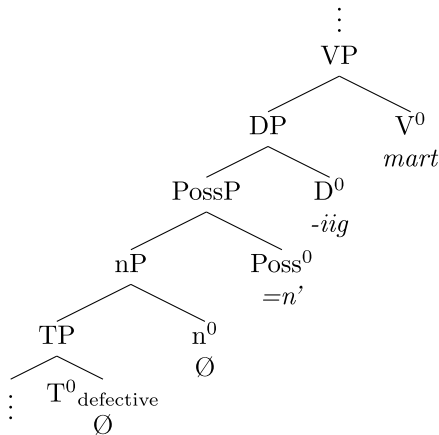


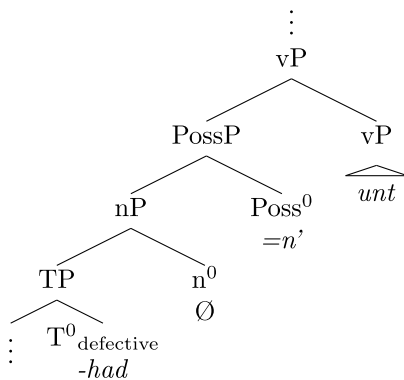
Table 1 Types of NCs in Khalkha

	PossP	DP
TP	Converbial adjunct	Nonfinite argument
CP	(Unattested)	Finite argument

(86) Nonfinite NC (80)



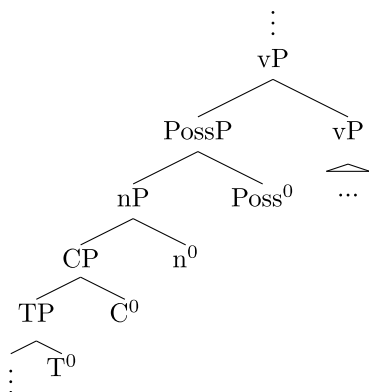
(87) Converbial adjunct (81)

**5.2 A gap in the typology of nominalized clauses**

Strikingly, if one considers all logically possible combinations of clausal and nominal structure, a gap in the data emerges, represented by the bottom left cell of Table 1 (repeated here). To the best of my knowledge, there is no evidence in Khalkha of a structure where a full CP is nominalized by a reduced PossP structure *lacking a DP structure over it*.

The unattested structure in Table 1 is represented in (88) as a type of adjunct clause adjoined to vP, similar to converbial adjuncts in Khalkha.

(88) Unattested mixed extended projection: (no DP) > PossP > nP > CP



Under the present account which takes there to be intermediate nominal projections between CP and DP, the missing structure (88) cannot be understood as a selectional restriction between two adjacent heads, for instance, that C^0 must be selected by D^0 . I argue that the gap is not accidental, and that it follows from more general principles of structure building.

5.3 Hierarchies of Domains

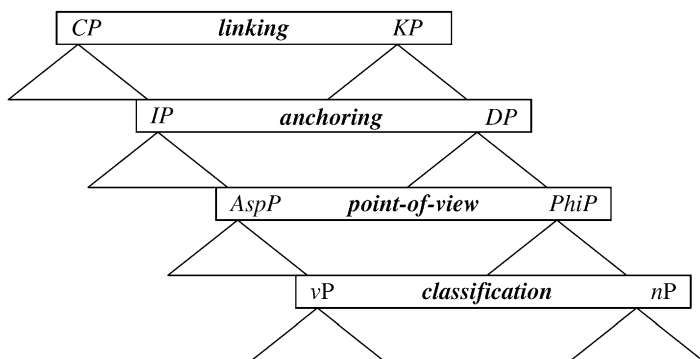
In fact, the gap resembles a prediction by Svenonius (2021), who proposes that mixed extended projections are constrained by what he calls Hierarchies of Domains (HoD):

HoD allows high nominal over high verbal (Spanish), high nominal over intermediate verbal (English gerund), or low nominal over low verbal (nominalizations like *destruction*), *but not low nominal over high verbal structure* [...] in which relatively high verbal morphology like tense or the perfect could be dominated by relatively low nominal characteristics...

(Svenonius 2021, 29; emphasis mine)

Svenonius' notion of HoD combines two ideas. The first idea is the common assumption that there exists a hierarchy of projections within the verbal and nominal extended projections (e.g., CP > TP > AspP > vP) and is familiar from works such as Abney (1987), Szabolcsi (1994), Rizzi (1997), Cinque (1999). The second idea comes from Wiltschko's (2014) Universal Spine Hypothesis, which posits that each verbal/nominal projection may be associated with a category-neutral domain along a more abstract functional spine, illustrated in (89). For instance, Wiltschko argues that the "anchoring" function is expressed by person-number marking in Blackfoot, while the same function is expressed by locative auxiliaries in Halkomelem (see also Ritter and Wiltschko 2014).

- (89) The parallelism between the nominal and verbal spine (Wiltschko 2014, 78)



The Universal Spine Hypothesis offers a framework to conceptualize parallels between the CP and DP domains, an idea also explored by Abney 1987; Grimshaw 1991; Giusti 2002, 2006; Hiraiwa 2005, among many others. For example, in (89), vP and nP belong to the same domain of ‘classification,’ AspP and PhiP belong to ‘point-of-view,’ and so on. Svenonius argues that it further constrains how domains combine: a verbal functional projection within a given domain cannot be dominated by a verbal or nominal functional projection from a lower domain. This restriction applies to extended and mixed extended projections but excludes, for example, complements or adjuncts. I restate Svenonius’ constraint as in (90):

- (90) HoD constraint on (mixed) extended projections

A verbal functional projection in domain α may not be dominated by a verbal/nominal functional projection in domain β , if $\alpha > \beta$.

(90) implies that a TP-sized clause belonging to the “anchoring” domain cannot be nominalized by a functional projection belonging to the lower domains of “point-of-view” and “classification”; it can only be nominalized by a nominal projection belonging to the “anchoring” (e.g., DP) or “linking” (e.g., KP (Wiltschko 2014) or PP (Svenonius 2021)) domain.^{35,36}

While Svenonius’s (2021) analysis is similar in spirit to my proposal outlined below, it does not appear to make correct predictions for the Khalkha data shown above, as it predicts that structures where nP or PossP is projected above CP—regardless of whether nP or PossP is dominated by DP or not—should be ill-formed. This is because n^0 and $Poss^0$, by virtue of them being lower than D^0 , would belong to lower functional domains than C^0 . However, in Khalkha finite NCs, nP, and PossP may be

³⁵In a similar vein, Alexiadou 2020 and Iordăchioaia 2020 argue that clausal structure containing at least TP must be directly nominalized by D^0 . For example, Iordăchioaia (2020) argues that, assuming DP is the counterpart of CP, one would not expect functional projections below DP(=CP) to be projected above TP. She further argues that given that AspP is located between vP and TP, AspP may be directly nominalized by DP or involve nominalizing nP.

³⁶Svenonius (2021, 13, fn.13) acknowledges that some flexibility must be allowed with regards to which projection maps onto which domain across languages. For example, as some languages contain CPs directly nominalized by D^0 , for instance in Ndebele (Pietraszko 2019), this suggests that D^0 and C^0 belong to the same domain in these languages, instead of K^0 and C^0 as seen in (89).

projected above CP *if* nP and PossP are dominated by DP. The unattested structure involves nP and PossP projected above CP *without* DP projected above those nominal layers. More generally, Svenonius's analysis is incompatible with my analysis of mixed extended projections where lower nominal projections must be projected *in addition to* DP, in order to account for the variation in nominal properties that we find across different NC types in Khalkha. The HoD constraint in (90) as it stands thus appears to be too restrictive.

5.4 Improper mixed extended projections

Building on Svenonius 2021's core insight, I propose instead that the relevant constraint is a constraint on the *maximal* projection of the nominal and clausal functional layers within a mixed extended projection (91).

(91) HoD constraint on mixed extended projections (revised)

Given a HoD where $\alpha > \beta$, a verbal/clausal functional projection whose *maximum of the extended projection* belongs to $v_\alpha P$ cannot be dominated by a nominal functional projection whose *maximum of the extended projection* belongs to $n_\beta P$.

I further argue that (91) can be understood in terms of the relative timings of clausal and nominal embedding, which indirectly restricts the relative size of the clausal and nominal extended projections in a mixed extended projection. My proposal builds on similar types of asymmetries that have been subsumed under the general heading of Williams Cycle phenomena, named after an original account of movement asymmetries proposed by Williams (2003). The basic intuition behind Williams' account is that a syntactic operation cannot take place between two sites, X and Y, if Y is higher than X in an assumed functional sequence (see, e.g., Starke 2001). Accordingly, such an operation is described as *improper*. While traditionally proposed to account for cases of improper movement (e.g., Williams 2003; Müller 2014), it has also been extended to account for parallel properties of agreement and case assignment, respectively called improper agreement (Keine 2020) and improper case (Poole 2023). In this vein, I shall refer to the illicit structure in (88) as an *improper mixed extended projection*.

Before discussing the constraint on mixed extended projections, it is useful to consider how Williams (2003) accounts for the impossibility of improper movement under a timing-based account of clausal embedding. Williams proposes the Level Embedding Conjecture (LEC), a constraint on the timing of clausal embedding, which indirectly relates to the size of the embedded clause. A version of the LEC as phrased in Poole 2023 is given in (92):

(92) Level Embedding Conjecture (LEC)

An XP can only be embedded in a structure that is built up to an XP.

The basic idea is that a clause of size XP can only be embedded within a clause of size XP or larger according to an assumed clausal functional sequence (see, e.g., Starke 2001). Under the LEC, both the matrix and embedded clauses are simultaneously built up in separate workspaces during the course of the derivation, which

can be thought of as comprising multiple stages along what Williams calls a “F(unctional)-clock.” For concreteness, I will follow Poole 2023 in adopting the use of a placeholder (PH) device and a substitution operation as a means to formalize how embedding works under the Williams Cycle. Additionally, the placeholder contains a “pointer” in the sense of Williams 2011 that specifies the construction that will be substituted for the placeholder; I will represent the pointer as a subscript to PH, for instance PH_{TP1} . Consider first the derivation of a licit structure [$_{CP}$ *Alex believes* [$_{TP}$ *Sam to be sick*]] under Level Embedding (vertical dashed lines are used to separate workspaces), shown in (93).

(93) Ruled in by the LEC: [$_{CP}$... [$_{TP}$...]]

	F-clock	Matrix root clause	Embedded clause 1
a.	F_{VP}	V_M	V_1
b.	$F_{VP(\text{insert PH})}$	$[V_M PH_{TP1}]$	$[V_1]$
c.	F_{TP}	$[T_M [V_M PH_{TP1}]]$	$[T_1 [V_1]]$
d.	$F_{TP(\text{substitution})}$	$[T_M [V_M [T_1 [V_1]]]]$	
e.	F_{CP}	$[C_M [T_M [V_M [T_1 [V_1]]]]]$	

In (93), the matrix clause and the embedded clause are simultaneously built up in separate workspaces, starting from the VP level (93a). At F_{VP} (93b), the matrix verb is merged with a placeholder PH_{TP1} that specifies TP1 as the construction it will be substituted for at the TP stage. At F_{TP} , both clauses are built up to the TP level (93c), following which the embedded TP1 is embedded into the matrix clause (93d). At F_{CP} (93e), the matrix clause, now containing the embedded TP1, is built up to the CP level.

Next, consider an illicit structure where a CP is embedded within a matrix root TP, for instance, $*[_{TP}$ *Alex to believe* [$_{CP}$ *that Sam is sick*]]. This is ruled out by the LEC as there is no grammatical derivation in which the embedded clause but not the matrix root clause has been built up to the CP level. However, such a structure is possible if the matrix TP is contained within a root CP, as in [$_{CP}$ *Taylor wanted* [$_{TP}$ *Alex to believe* [$_{CP}$ *that Sam is sick*]]]. Consider (94) below.

(94) Ruled in by the LEC: [$_{CP}$... [$_{TP}$... [$_{CP}$...]]]

	F-clock	Matrix root clause	Emb. clause 1	Emb. clause 2
a.	F_{VP}	V_M	V_1	V_2
b.	$F_{VP(\text{ins. PH})}$	$[V_M PH_{TP1}]$	$[V_1 PH_{CP2}]$	$[V_2]$
c.	F_{TP}	$[T_M [V_M PH_{TP1}]]$	$[T_1 [V_1 PH_{CP2}]]$	$[T_2 [V_2]]$
d.	$F_{TP(\text{sub.})}$	$[T_M [V_M [T_1 [V_1 PH_{CP2}]]]]$		$[T_2 [V_2]]$
e.	F_{CP}	$[C_M [T_M [V_M [T_1 [V_1 PH_{CP2}]]]]]$		$[C_2 [T_2 [V_2]]]$
f.	$F_{CP(\text{sub.})}$	$[C_M [T_M [V_M [T_1 [V_1 [C_2 [T_2 [V_2]]]]]]]]]$		

In (94), the middle TP contains a placeholder that will be substituted with the most embedded CP. At F_{TP} (94d), the middle TP is embedded into the root clause.

The most embedded clause continues to be built in parallel with the root clause up to CP at F_{CP} (94e). Finally, the most embedded clause is embedded into the matrix root clause, as shown in (94f). Essentially, a CP can be embedded into a TP if, at the time of embedding, the TP is already in the matrix root clause. Thus, the LEC does not block any specific type of embedding, for instance $TP > CP$, but dictates the *timing* of the embedding (which is indirectly related to the size of the embedding).

Going back to NCs, I assume that some form of parallelism between the CP and DP domains obtains at a more general or abstract level, for instance, the categorial parallelism posited by Hiraiwa (2005) or the functional parallelism under Wiltschko's (2014) Universal Spine Hypothesis. I adopt Svenonius's (2021) idea that the parallelism between the CP and DP domains can be mapped onto a HoD. For concreteness, I propose that Khalkha possesses the HoD in (95), where d stands for an abstract domain to which a particular clausal and nominal projection is mapped. Note that some clausal projections (e.g., VoiceP, AspP) are not represented in (95) but may be mapped to other nominal projections (e.g., NumP) under a more comprehensive conception of the HoD in Khalkha.

(95) HoD in Khalkha

	d_1	d_2	d_3	d_4
Clausal	V	v	T	C
Nominal	N	n	Poss	D

Once the idea of a HoD like (95) is accepted, it is predicted that not only is nominal structure building subject to the LEC, but also that any interaction between clausal and nominal layers will be subject to the LEC. For instance, one would predict that a clausal layer may only be embedded under a nominal layer belonging to a parallel or higher domain, but not under a nominal layer belonging to a lower domain.

Williams (2011, 2013) also further differentiates nominal embedding from clausal embedding. While clausal embedding occurs through Level Embedding, nominal embedding operates via Co-generation, where a verbal category is merged with a placeholder to be replaced by a nominal construction. Notably, clausal embedding takes place once the embedded clause reaches the size required by the embedding predicate, whereas nominal embedding occurs *only at the end of the derivation*, for instance, at F_{CP} . Following Williams, I assume that once a structure is embedded (i.e., substituted for its placeholder), no additional structure can be built above it, as only top-level elements in a single workspace (i.e., the outermost nominal or clausal layers) are eligible for further derivation (Williams 2011, 164). Additionally, at F_{VP} , I assume that NP is not constructed within the nominal layer of Khalkha NCs (as Khalkha NCs are not N-CP structures); instead, a placeholder is inserted with a pointer indicating the size of the clausal structure to be embedded under the nominal layer.

With this background in mind, consider a licit derivation of an embedded nominalized clause, a finite NC in Khalkha with the structure $[CP \dots [DP \dots [CP \dots]]]$. This is shown in (96).

(96) Ruled in by Co-generation and LEC: $[_{CP} \dots [_{DP} \dots [_{CP} \dots]]]$

	F-clock	Matrix root clause	Embedded NC	
			Nominal layer 1	Clause 2
a.	F_{VP}	V_M		V_2
b.	$F_{VP(ins. PH)}$	$[V_M PH_{DP1}]$		V_2
c.	F_{vP}	$[v_M [V_M PH_{DP1}]]$	$[n_1]$	$[v_2 V_2]$
d.	$F_{vP(ins. PH)}$	$[v_M [V_M PH_{DP1}]]$	$[n_1 PH_{CP2}]$	$[v_2 V_2]$
e.	F_{TP}	$[T_M [v_M [V_M PH_{DP1}]]]$	$[Poss_1 [n_1 PH_{CP2}]]$	$[T_2 [v_2 V_2]]$
f.	F_{CP}	$[C_M [T_M [v_M [V_M PH_{DP1}]]]]$	$[D_1 [Poss_1 [n_1 PH_{CP2}]]]$	$[C_2 [T_2 [v_2 V_2]]]$
g.	$F_{CP(sub.)}$	$[C_M [T_M [v_M [V_M PH_{DP1}]]]]$	$[D_1 [Poss_1 [n_1 [C_2 [T_2 [v_2 V_2]]]]]]$	
h.	$F_{CP(sub.)}$	$[C_M [T_M [v_M [V_M [D_1 [Poss_1 [n_1 [C_2 [T_2 [v_2 V_2]]]]]]]]]$		

At F_{VP} (96a), both the matrix root clause and the most embedded clause (Clause 2) are built up to VP; NP is not built within the nominal layer. Next, as shown in (96b), a placeholder PH_{DP1} is inserted into the matrix root VP. At (96c), all three layers continue to be built up to vP and nP. At (96d), PH_{CP2} is inserted into the nominal layer of the embedded NC, specifying that Clause 2 is to be embedded into the nominal layer when it has been built to CP. All three layers continue to be built up to TP/PossP and CP/DP at F_{TP} (96e) and F_{CP} (96f). After all three structures have been built up to CP or DP, two embedding operations occur. First, as shown in (96g), Clause 2 has reached the size required by the nominal layer and it substitutes PH_{CP2} via Level Embedding. Second, in the final step of the derivational cycle (96h), the entire nominal layer containing Clause 2 is embedded into the matrix root clause via Co-generation.

For a nonfinite NC with the structure $[_{CP} \dots [_{DP} \dots [_{TP} \dots]]]$, the middle nominal layer contains PH_{TP2} instead of PH_{CP2} . At F_{TP} , the most embedded clause, built up to TP_2 , embeds into the middle nominal layer. At F_{CP} , the matrix root clause and the middle nominal layer are built up to CP and DP, respectively. In the final substitution step, the nominal layer, which already contains TP_2 , is embedded into the matrix root clause.

Essentially, once some form of parallelism between the clausal and nominal domains is assumed, the logics of clausal and nominal embedding under the Williams Cycle derive the revised HoD constraint (91), repeated below:

(91) *HoD constraint on mixed extended projections (revised)*

Given a HoD where $\alpha > \beta$, a verbal/clausal functional projection whose *maximum of the extended projection* belongs to $v_\alpha P$ cannot be dominated by a nominal functional projection whose *maximum of the extended projection* belongs to $n_\beta P$.

I illustrate how (91) can be derived via the derivation of a SNC under the Williams Cycle. (97) shows a converbial adjunct embedded in a matrix root clause, which has the structure $[_{CP} \dots [_{PossP} \dots [_{TP} \dots]]]$. Since converbial adjuncts adjoin to vP, I propose that the placeholder for converbial adjuncts— PH_{PossP} —is inserted at F_{vP} .

- (97) Ruled in by Co-generation and LEC: $[CP \dots [PossP \dots [TP \dots]]]$
 Ruled out by LEC: $[CP \dots [PossP \dots [CP \dots]]]$

	F-clock	Matrix root clause	Embedded NC	
			Nominal layer 1	Clause 2
a.	F_{VP}	V_M		V_2
b.	F_{VP}	$[V_M V_M]$	$[n_1]$	$[v_2 V_2]$
c.	$F_{VP(ins. PH)}$	$[PH_{PossP1} [V_M V_M]]$	$[n_1 PH_{TP2}]$	$[v_2 V_2]$
d.	F_{TP}	$[T_M [PH_{PossP1} [V_M V_M]]]$	$[Poss_1 [n_1 PH_{TP2}]]$	$[T_2 [v_2 V_2]]$
e.	$F_{TP(sub.)}$	$[T_M [PH_{PossP1} [V_M V_M]]]$	$[Poss_1 [n_1 [T_2 [v_2 V_2]]]]$	
f.	F_{CP}	$[C_M [T_M [PH_{PossP1} [V_M V_M]]]]$	$[Poss_1 [n_1 [T_2 [v_2 V_2]]]]$	
g.	$F_{CP(sub.)}$	$[C_M [T_M [Poss_1 [n_1 [T_2 [v_2 V_2]]]] [V_M V_M]]]$	$\uparrow$ <i>CP cannot be built</i>	

At F_{VP} (97a), VP is built in both the matrix root clause and Clause 2. At F_{VP} (97b), vP is built in the matrix root clause and Clause 2, and nP is built in the middle nominal layer. Next, as seen in (97c), two placeholders are inserted: PH_{PossP1} is inserted into the matrix root clause, specifying that $PossP_1$ will embed into the matrix root clause at the end of the derivation; and PH_{TP2} is inserted into the middle nominal layer, indicating that Clause 2 will be embedded under the nominal layer once it reaches the size of TP. At F_{TP} (97d), TP is constructed in both the matrix root clause and Clause 2, while $PossP$ is built in the middle nominal layer. The first substitution occurs in (97e) as Clause 2 reaches the required size and substitutes for PH_{TP2} in the middle nominal layer due to Level Embedding. At F_{CP} (97f), only the CP of the matrix root clause is constructed. Finally, as seen in (97g), $PossP_1$ is embedded into the matrix root clause.

Consider what it would take to generate an improper mixed extended projection where the middle nominal layer— $PossP$ —contains a CP-sized clause. One possibility is that CP may be built above TP after the most embedded clause—Clause 2—is embedded into the nominal layer. However, as shown in (97f), this is ruled out as only top-level structures within a single workspace can have additional structure built above them.

Another possibility is to delay the substitution of the clause into the nominal layer until after CP has been built at F_{CP} . However, this would run afoul of the LEC, as shown in (98f). Essentially, at the point where the most embedded clause CP_2 is to be embedded into the middle nominal layer (98f), the LEC dictates that it must be of an equal size to CP (i.e. DP). If the nominal layer were already embedded into the matrix root clause, which has been built up to CP, this would satisfy the LEC. However, this is ruled out by Co-generation: because nominals are only embedded at the end of the derivation, the size of the matrix root clause is not relevant at (98f). Since the nominal layer has only been built up to $PossP$ at (98f), embedding CP_2 into $PossP$ violates LEC and is ruled out.

Table 7 (Im)possible nominalized structures

Nominal layer	Clausal structure			
	VP	vP	TP	CP
nP	✓	✓	✗	✗
PossP	✓	✓	✓	✗
DP	✓	✓	✓	✓

(98) Ruled out by LEC : [CP ... [PossP ... [CP ...]]]

F-clock	Matrix root clause	Embedded NC	
		Nominal layer 1	Clause 2
a. F _{VP}	V _M		V ₂
b. F _{VP}	[V _M V _M]	[n ₁]	[v ₂ V ₂]
c. F _{VP (ins. PH)}	[PH _{PossP1} [V _M V _M]]	[n ₁ PH _{CP2}]	[v ₂ V ₂]
d. F _{TP}	[T _M [PH _{PossP1} [V _M V _M]]]	[Poss ₁ [n ₁ PH _{CP2}]]	[T ₂ [v ₂ V ₂]]
e. F _{CP}	[C _M [T _M [PH _{PossP1} [V _M V _M]]]]	[Poss ₁ [n ₁ PH _{CP2}]]	[C ₂ [T ₂ [v ₂ V ₂]]]
f. F _{CP (sub.)}	[C _M [T _M [PH _{PossP1} [V _M V _M]]]]	[Poss ₁ [n ₁ PH _{CP2}]]	[C ₂ [T ₂ [v ₂ V ₂]]]

$\uparrow$ ✗
 LEC violation

In addition to improper mixed extended projections of the type illustrated above, embedding under the Williams Cycle also rules out *any* mixed extended projection where the clausal structure's maximal projection belongs to a higher abstract domain than the nominal structure's maximal projection, predicted by (91). For example, in a language with a HoD like Khalkha, a CP can be dominated by a nominal superstructure only if the superstructure contains DP, a TP can be dominated by a nominal superstructure only if the superstructure contains PossP, and so forth. The resulting typology of possible and impossible nominalized structures is illustrated in Table 7.

5.5 Summary

To summarize, in this section, I have discussed a gap in the empirical landscape of Khalkha NCs: the absence of a structure in which a full CP is nominalized by reduced PossP. I have shown that this unattested structure, which I term an improper mixed extended projection, basically falls out under Williams' (2003, 2011, 2013) ideas about clausal and nominal embedding once some form of parallelism between the clausal and nominal domain (e.g., Wiltschko 2014) is assumed. The present account also derives a revised version of Svenonius's (2021) HoD constraint on mixed extended projections. Finally, the present account predicts that there should be no nominalized structures in which the maximal projection within the nominal exterior belongs to a lower level in a given HoD than the maximal projection of the clausal structure being nominalized.

However, recall from the Sect. 1 that Nasu et al. (2024) argue that Japanese contains NCs in which a reduced nP lacking a DP layer dominates a full CP. This is shown in (99).

(99) Japanese

- a. John-wa [(**ano*) nani-o Mary-ga katta ka]-o {tazuneta/touta}.
 J.-TOP that what-ACC M. bought Q-ACC asked/questioned
 ‘John asked/questioned what Mary had bought.’
- b. Kyoo-no kaigi-de [*ano* dooyatte kakaku-o osaeru ka]-ga
 today-GEN meeting-IN that how price-ACC control Q-NOM
 hanasiaw-are-ta.
 discuss-PASS-PST
 ‘That how we control the price was discussed in today’s meeting’

(Nasu et al. 2024, 13, exx. 28–29)

Both embedded clauses (99a) and (99b) contain the question particle *ka*, suggesting that they contain a full CP structure. Both embedded clauses are also marked with case, suggesting that they are nominalized. Nasu et al. (2024) argue that the anaphoric demonstrative *ano* ‘that’ occupies SpecDP in Japanese. Since (99a) may not contain *ano* but (99b) may do so, the authors argue that the NC in (99a) involves a full CP dominated by a nP exterior without a DP layer, while the NC in (99b) involves a full CP dominated by a nominal exterior containing nP and DP.

However, the data presented by Nasu et al. (2024) also exhibit some unexplained properties that warrant further investigation. First, the presence of case marking on the embedded clause in (99a) suggests that there may be additional nominal functional structure above nP within the nominal superstructure above the clause. Second, for clauses that may contain the demonstrative *ano*, the authors only show subject clauses marked with nominative case *-ga* (e.g., (99b)); for clauses that may not contain the demonstrative *ano*, the authors only show complement clauses marked with accusative case *-o* (e.g., (99a)). Given that subject clauses are commonly noted to contain more nominal structure than complement clauses cross-linguistically (see, e.g., Davies and Dubinsky 1998, 2001, 2009; Hartman 2012), it may not be surprising that subject clauses in Japanese may contain an overt demonstrative while complement clauses may not. For example, Pietraszko (2019) notes that subject clauses in Polish obligatorily contain a demonstrative, while complement clauses may not contain one. Pending further investigation and given that there may be other ways to think about the Japanese data in (99b), I will assume that the Japanese data does not constitute a counterexample to the central claim advanced in this section: that a smaller nominal layer may not embed a larger clausal layer.

6 Conclusion

In this paper, I have shown that Khalkha NCs may vary in two key aspects. In terms of clausal structure, they may contain a full CP, which correlates with the presence of an overt complementizer and finite tense morphology, or they may lack CP structure, which correlates with the lack of an overt complementizer and a lack of finite tense morphology. In terms of nominal structure, they may be full DPs, paralleling the morphosyntactic properties of possessive nominals, or they may be reduced Pos-sPs, thus sharing only a subset of nominal properties with possessive nominals. In

showing that converbial adjuncts contain a reduced PossP nominal structure, I have shown how the full DP-SN distinction can also be observed in the clausal domain, even though this distinction is not observed in the nominal domain of Khalkha. This paper thus offers a case study on how to look for SNs in a language which may not straightforwardly exhibit reduced nominal structures in the nominal domain.

The question remains as to why there is an asymmetry in Khalkha as to where SNs can and cannot be found. As seen in Russian and Tatar, SNs can serve as arguments of verbs. As Khalkha does not seem to contain argument small nominals that are not (pseudo-)incorporated, one would have to look at nonarguments, for instance adjuncts, which cannot be simplex nominals in the language and only appear as clause-sized constituents. This is why the full DP-SN distinction maps onto the argument-adjunct clause distinction in Khalkha. A prediction here is that a language that allows argument small nominals may also allow argument SNCs, if nominalized clauses are possible in that language to begin with. If this turns out to be correct, then it suggests that the argument-adjunct split between full DPs and SNCs in Khalkha is not due to a universal constraint pertaining to the argument vs. adjunct distinction, but a mere coincidence owing to the lack of non-(pseudo-)incorporated SN arguments. It also suggests that adjunct small nominals may be found in a language where simplex nominals could function as adverbials.

Finally, I have proposed a typology of possible and impossible nominalized structures based on Williams' (2003, 2011, 2013) theory of clausal and nominal embedding, which derives a revised version of Svenonius's (2021) HoD constraint on building mixed extended projections. This proposed typology further yields two interesting predictions. First, assuming that languages that do not contain DPs in the purely nominal domain also do not contain DP-sized nominalizations, the proposed typology predicts that such languages will not be able to nominalize CPs. This may be true of so-called NP-languages, which have been argued to lack DP-sized nominals (Bošković 2008, et seq.). This requires the further assumption that CPs in NP-languages completely lack a parallel in the nominal domain.³⁷ Second, the proposed typology predicts that if a language contains only CP-sized embedded clauses, wholly lacking reduced embedded clauses, in such a language we will not find SNCs.

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Declarations

Competing Interests The author has no conflict of interest to declare.

³⁷ Alternatively, it may be the case that the highest non-DP nominal projection serves as the parallel for CP in a NP-language. I leave this issue for future research.

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